```
Unknown command 'clear' - try 'help'
STM32MP> mmc dev 1
switch to partitions #0, OK
mmc1(part 0) is current device
STM32MP> fatload mmc 1:2 ${kernel_addr_r} uImage
8171336 bytes read in 315 ms (24.7 MiB/s)
STM32MP> fatload mmc 1:2 ${fdt_addr_r} stm32mp157c-100ask-512d-v1.dtb
73271 bytes read in 40 ms (1.7 MiB/s)
STM32MP> bootm ${kernel_addr_r} - ${fdt_addr_r}
## Booting kernel from Legacy Image at c2000000 ...
   Image Name:   Linux-5.4.31-g893d15a84
   Created:      2026-10-21  11:38:18 UTC
   Image Type:   ARM Linux Kernel Image (uncompressed)
   Data Size:    8171272 Bytes = 7.8 MiB
   Load Address: 0c200040
   Entry Point:  0c200040
   Verifying Checksum ... OK
## Flattened Device Tree blob at c4000000
   Booting using the fdt blob at 0xc4000000
   Loading Kernel Image
   Loading Device Tree to cffeb000, end cffffe36 ... OK

Starting kernel ...

prefetch abort
pc : [<0c20003c>]          lr : [<d9c47573>]
reloc pc : [<f26ba03c>]    lr : [<c0101573>]
sp : d7aec8c0  ip : d7c45102     fp : c2000000
r10: d9c47695  r9 : d7affec0     r8 : d7c4da08
r7 : 00000000  r6 : 0c200040     r5 : d9cf4030  r4 : 00000000
r3 : 00011e37  r2 : cffeb000     r1 : 00000000  r0 : 00000000
Flags: nZCv  IRQs off  FIQs off  Mode SVC_32
Code: data abort
pc : [<d9c47920>]          lr : [<d9c4790d>]
reloc pc : [<c0101920>]    lr : [<c010190d>]
sp : d7aec7c8  ip : 0000003c     fp : c2000000
r10: d9cd478a  r9 : d7affec0     r8 : d9cd4782
r7 : 00000000  r6 : 00000008     r5 : 0c20003c  r4 : fffffffc
r3 : 00000001  r2 : 00000028     r1 : 00000000  r0 : 00000006
Flags: NzCv  IRQs on  FIQs on  Mode SVC_32 (T)
Code: 2f00 d1e9 2c00 dce9 (f855) 2024
Resetting CPU ...

resetting ...
INFO:    PSCI Power Domain Map:
INFO:      Domain Node : Level 1, parent_node -1, State ON (0x0)
INFO:      Domain Node : Level 0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0x0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0xffffffff, parent_node 0, State OFF (0x2)
NOTICE:  CPU: STM32MP157AAC Rev.Z
NOTICE:  Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
INFO:    Reset reason (0x54):
INFO:      System reset generated by MPU (MPSYSRST)
```

```
INFO:    Using EMMC
INFO:      Instance 2
INFO:    Boot used partition fsbl1
NOTICE:  BL2: v2.2-r1.0(debug):85b7ab8
NOTICE:  BL2: Built : 09:15:17, Aug 16 2026
INFO:    Using crypto library 'stm32_crypto_lib'
INFO:    BL2: Doing platform setup
INFO:    RAM: DDR3-1066/888 bin G 2x2Gb 533MHz v1.44
INFO:    Memory size = 0x20000000 (512 MB)
INFO:    BL2 runs SP_MIN setup
INFO:    BL2: Loading image id 4
INFO:    Loading image id=4 at address 0x2ffed000
INFO:    Image id=4 loaded: 0x2ffed000 - 0x2ffff000
INFO:    BL2: Loading image id 5
INFO:    Loading image id=5 at address 0xc0100000
INFO:    STM32 Image size : 869897
INFO:    Image id=5 loaded: 0xc0100000 - 0xc01d4609
WARNING: Skip signature check (header option)
NOTICE:  ROTPK is not deployed on platform. Skipping ROTPK verification.
NOTICE:  BL2: Booting BL32
INFO:    Entry point address = 0x2ffed000
INFO:    SPSR = 0x1d3
NOTICE:  SP_MIN: v2.2-r1.0(debug):85b7ab8
NOTICE:  SP_MIN: Built : 09:15:17, Aug 16 2026
INFO:    ARM GICv2 driver initialized
INFO:    stm32mp IWDG1 (12): Secure
INFO:    ETZPC: CRYP1 (9) could be non secure
INFO:    SP_MIN: Initializing runtime services
INFO:    SP_MIN: Preparing exit to normal world


U-Boot 2020.01-stm32mp-r1 (Aug 16 2026 - 09:15:35 +0800)

CPU: STM32MP157AAC Rev.Z
Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
Board: stm32mp1 in trusted mode (st,stm32mp157c-100ask-512d-v1)
DRAM:  512 MiB
Clocks:
- MPU : 650 MHz
- MCU : 208.878 MHz
- AXI : 266.500 MHz
- PER : 24 MHz
- DDR : 533 MHz
WDT:   Started with servicing (32s timeout)
NAND:  0 MiB
MMC:   STM32 SD/MMC: 0, STM32 SD/MMC: 1
Loading Environment from MMC... OK
In:    serial
Out:   serial
Err:   serial
Net:   eth0: ethernet@5800a000
Press SPACE to abort autoboot in 1 seconds
Boot over mmc1!
switch to partitions #0, OK
```

```
mmc1(part 0) is current device
Scanning mmc 1:2...
Found /mmc1_extlinux/stm32mp157c-100ask-512d-v1_extlinux.conf
Retrieving file: /mmc1_extlinux/stm32mp157c-100ask-512d-v1_extlinux.conf
210 bytes read in 38 ms (4.9 KiB/s)
1:      100ask-lcd
Retrieving file: /uImage
8171336 bytes read in 316 ms (24.7 MiB/s)
append: root=/dev/mmcblk2p3 rootwait rw console=ttySTM0,115200
Retrieving file: /stm32mp157c-100ask-512d-lcd-v1.dtb
74438 bytes read in 40 ms (1.8 MiB/s)
## Booting kernel from Legacy Image at c2000000 ...
   Image Name:   Linux-5.4.31-g893d15a84
   Created:      2026-10-21  11:38:18 UTC
   Image Type:   ARM Linux Kernel Image (uncompressed)
   Data Size:    8171272 Bytes = 7.8 MiB
   Load Address: 0c200040
   Entry Point:  0c200040
   Verifying Checksum ... OK
## Flattened Device Tree blob at c4000000
   Booting using the fdt blob at 0xc4000000
   Loading Kernel Image
   Loading Device Tree to cffea000, end cffff2c5 ... OK

Starting kernel ...

prefetch abort
pc : [<0c20003c>]          lr : [<d9c47573>]
reloc pc : [<f26ba03c>]    lr : [<c0101573>]
sp : d7aeb278  ip : d7c45102     fp : c2000000
r10: d9c47695  r9 : d7affec0     r8 : d7c5a628
r7 : 00000000  r6 : 0c200040     r5 : d9cf4030  r4 : 00000000
r3 : 000122c6  r2 : cffea000     r1 : 00000000  r0 : 00000000
Flags: nZCv  IRQs off  FIQs off  Mode SVC_32
Code: data abort
pc : [<d9c47920>]          lr : [<d9c4790d>]
reloc pc : [<c0101920>]    lr : [<c010190d>]
sp : d7aeb180  ip : 0000003c     fp : c2000000
r10: d9cd478a  r9 : d7affec0     r8 : d9cd4782
r7 : 00000000  r6 : 00000008     r5 : 0c20003c  r4 : fffffffc
r3 : 00000001  r2 : 00000028     r1 : 00000000  r0 : 00000006
Flags: NzCv  IRQs on  FIQs on  Mode SVC_32 (T)
Code: 2f00 d1e9 2c00 dce9 (f855) 2024
Resetting CPU ...

resetting ...
INFO:    PSCI Power Domain Map:
INFO:      Domain Node : Level 1, parent_node -1, State ON (0x0)
INFO:      Domain Node : Level 0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0x0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0xffffffff, parent_node 0, State OFF (0x2)
NOTICE:  CPU: STM32MP157AAC Rev.Z
NOTICE:  Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
INFO:    Reset reason (0x54):
```

```
INFO:      System reset generated by MPU (MPSYSRST)
INFO:    Using EMMC
INFO:      Instance 2
INFO:    Boot used partition fsbl1
NOTICE:  BL2: v2.2-r1.0(debug):85b7ab8
NOTICE:  BL2: Built : 09:15:17, Aug 16 2026
INFO:    Using crypto library 'stm32_crypto_lib'
INFO:    BL2: Doing platform setup
INFO:    RAM: DDR3-1066/888 bin G 2x2Gb 533MHz v1.44
INFO:    Memory size = 0x20000000 (512 MB)
INFO:    BL2 runs SP_MIN setup
INFO:    BL2: Loading image id 4
INFO:    Loading image id=4 at address 0x2ffed000
INFO:    Image id=4 loaded: 0x2ffed000 - 0x2ffff000
INFO:    BL2: Loading image id 5
INFO:    Loading image id=5 at address 0xc0100000
INFO:    STM32 Image size : 869897
INFO:    Image id=5 loaded: 0xc0100000 - 0xc01d4609
WARNING: Skip signature check (header option)
NOTICE:  ROTPK is not deployed on platform. Skipping ROTPK verification.
NOTICE:  BL2: Booting BL32
INFO:    Entry point address = 0x2ffed000
INFO:    SPSR = 0x1d3
NOTICE:  SP_MIN: v2.2-r1.0(debug):85b7ab8
NOTICE:  SP_MIN: Built : 09:15:17, Aug 16 2026
INFO:    ARM GICv2 driver initialized
INFO:    stm32mp IWDG1 (12): Secure
INFO:    ETZPC: CRYP1 (9) could be non secure
INFO:    SP_MIN: Initializing runtime services
INFO:    SP_MIN: Preparing exit to normal world


U-Boot 2020.01-stm32mp-r1 (Aug 16 2026 - 09:15:35 +0800)

CPU: STM32MP157AAC Rev.Z
Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
Board: stm32mp1 in trusted mode (st,stm32mp157c-100ask-512d-v1)
DRAM:  512 MiB
Clocks:
- MPU : 650 MHz
- MCU : 208.878 MHz
- AXI : 266.500 MHz
- PER : 24 MHz
- DDR : 533 MHz
WDT:   Started with servicing (32s timeout)
NAND:  0 MiB
MMC:   STM32 SD/MMC: 0, STM32 SD/MMC: 1
Loading Environment from MMC... OK
In:    serial
Out:   serial
Err:   serial
Net:   eth0: ethernet@5800a000
Press SPACE to abort autoboot in 1 seconds
STM32MP> mmc dev 1
```

```
switch to partitions #0, OK
mmc1(part 0) is current device
STM32MP> fatload mmc 1:2 0xc200040 uImage
** Reading file would overwrite reserved memory **
STM32MP> fatload mmc 1:2 0xc4000000 stm32mp157c-100ask-512d-v1.dtb
73271 bytes read in 41 ms (1.7 MiB/s)
STM32MP> setenv bootargs 'root=/dev/mmcblk1p3 rootwait rw console=ttySTM0,115
200'
STM32MP> bootm ${kernel_addr_r} - ${fdt_addr_r}
Wrong Image Format for bootm command
ERROR: can't get kernel image!
STM32MP>
```

拒绝访问。

```
Connection closed.

Disconnected from remote host(MP157) at 17:19:47.

Type `help' to learn how to use Xshell prompt.
[C:\~]$

Connecting to COM5...
Failed to open COM5.

Type `help' to learn how to use Xshell prompt.
[C:\~]$

Connecting to COM5...
Connected.


Wrong Image Format for bootm command
ERROR: can't get kernel image!
STM32MP> mmc dev 1
switch to partitions #0, OK
mmc1(part 0) is current device
STM32MP> fatload mmc 1:2 0xc4200000 uImage
8171336 bytes read in 315 ms (24.7 MiB/s)
STM32MP> fatload mmc 1:2 0xc4000000 stm32mp157c-100ask-512d-v1.dtb
73271 bytes read in 40 ms (1.7 MiB/s)
STM32MP> setenv bootargs 'root=/dev/mmcblk1p3 rootwait rw console=ttySTM0,115
200'
STM32MP> bootm 0xc4200000 - 0xc4000000
## Booting kernel from Legacy Image at c4200000 ...
   Image Name:   Linux-5.4.31-g893d15a84
   Created:      2026-10-21  11:38:18 UTC
   Image Type:   ARM Linux Kernel Image (uncompressed)
   Data Size:    8171272 Bytes = 7.8 MiB
   Load Address: 0c200040
   Entry Point:  0c200040
   Verifying Checksum ... OK
## Flattened Device Tree blob at c4000000
   Booting using the fdt blob at 0xc4000000
```

```
   Loading Kernel Image
   Loading Device Tree to cffeb000, end cffffe36 ... OK

Starting kernel ...

prefetch abort
pc : [<0c20003c>]          lr : [<d9c47573>]
reloc pc : [<f26ba03c>]    lr : [<c0101573>]
sp : d7aec9a0  ip : d7c45102     fp : c4200000
r10: d9c47695  r9 : d7affec0     r8 : d7c4d5c8
r7 : 00000000  r6 : 0c200040     r5 : d9cf4030  r4 : 00000000
r3 : 00011e37  r2 : cffeb000     r1 : 00000000  r0 : 00000000
Flags: nZCv  IRQs off  FIQs off  Mode SVC_32
Code: data abort
pc : [<d9c47920>]          lr : [<d9c4790d>]
reloc pc : [<c0101920>]    lr : [<c010190d>]
sp : d7aec8a8  ip : 0000003c     fp : c4200000
r10: d9cd478a  r9 : d7affec0     r8 : d9cd4782
r7 : 00000000  r6 : 00000008     r5 : 0c20003c  r4 : fffffffc
r3 : 00000001  r2 : 00000028     r1 : 00000000  r0 : 00000006
Flags: NzCv  IRQs on  FIQs on  Mode SVC_32 (T)
Code: 2f00 d1e9 2c00 dce9 (f855) 2024
Resetting CPU ...

resetting ...
INFO:    PSCI Power Domain Map:
INFO:      Domain Node : Level 1, parent_node -1, State ON (0x0)
INFO:      Domain Node : Level 0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0x0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0xffffffff, parent_node 0, State OFF (0x2)
NOTICE:  CPU: STM32MP157AAC Rev.Z
NOTICE:  Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
INFO:    Reset reason (0x54):
INFO:      System reset generated by MPU (MPSYSRST)
INFO:    Using EMMC
INFO:      Instance 2
INFO:    Boot used partition fsbl1
NOTICE:  BL2: v2.2-r1.0(debug):85b7ab8
NOTICE:  BL2: Built : 09:15:17, Aug 16 2026
INFO:    Using crypto library 'stm32_crypto_lib'
INFO:    BL2: Doing platform setup
INFO:    RAM: DDR3-1066/888 bin G 2x2Gb 533MHz v1.44
INFO:    Memory size = 0x20000000 (512 MB)
INFO:    BL2 runs SP_MIN setup
INFO:    BL2: Loading image id 4
INFO:    Loading image id=4 at address 0x2ffed000
INFO:    Image id=4 loaded: 0x2ffed000 - 0x2ffff000
INFO:    BL2: Loading image id 5
INFO:    Loading image id=5 at address 0xc0100000
INFO:    STM32 Image size : 869897
INFO:    Image id=5 loaded: 0xc0100000 - 0xc01d4609
WARNING: Skip signature check (header option)
NOTICE:  ROTPK is not deployed on platform. Skipping ROTPK verification.
NOTICE:  BL2: Booting BL32
```

```
INFO:    Entry point address = 0x2ffed000
INFO:    SPSR = 0x1d3
NOTICE:  SP_MIN: v2.2-r1.0(debug):85b7ab8
NOTICE:  SP_MIN: Built : 09:15:17, Aug 16 2026
INFO:    ARM GICv2 driver initialized
INFO:    stm32mp IWDG1 (12): Secure
INFO:    ETZPC: CRYP1 (9) could be non secure
INFO:    SP_MIN: Initializing runtime services
INFO:    SP_MIN: Preparing exit to normal world


U-Boot 2020.01-stm32mp-r1 (Aug 16 2026 - 09:15:35 +0800)

CPU: STM32MP157AAC Rev.Z
Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
Board: stm32mp1 in trusted mode (st,stm32mp157c-100ask-512d-v1)
DRAM:  512 MiB
Clocks:
- MPU : 650 MHz
- MCU : 208.878 MHz
- AXI : 266.500 MHz
- PER : 24 MHz
- DDR : 533 MHz
WDT:   Started with servicing (32s timeout)
NAND:  0 MiB
MMC:   STM32 SD/MMC: 0, STM32 SD/MMC: 1
Loading Environment from MMC... OK
In:    serial
Out:   serial
Err:   serial
Net:   eth0: ethernet@5800a000
Press SPACE to abort autoboot in 1 seconds
Boot over mmc1!
switch to partitions #0, OK
mmc1(part 0) is current device
Scanning mmc 1:2...
Found /mmc1_extlinux/stm32mp157c-100ask-512d-v1_extlinux.conf
Retrieving file: /mmc1_extlinux/stm32mp157c-100ask-512d-v1_extlinux.conf
210 bytes read in 38 ms (4.9 KiB/s)
1:      100ask-lcd
Retrieving file: /uImage
8171336 bytes read in 315 ms (24.7 MiB/s)
append: root=/dev/mmcblk2p3 rootwait rw console=ttySTM0,115200
Retrieving file: /stm32mp157c-100ask-512d-lcd-v1.dtb
74438 bytes read in 40 ms (1.8 MiB/s)
## Booting kernel from Legacy Image at c2000000 ...
   Image Name:   Linux-5.4.31-g893d15a84
   Created:      2026-10-21  11:38:18 UTC
   Image Type:   ARM Linux Kernel Image (uncompressed)
   Data Size:    8171272 Bytes = 7.8 MiB
   Load Address: 0c200040
   Entry Point:  0c200040
   Verifying Checksum ... OK
## Flattened Device Tree blob at c4000000
```

```
   Booting using the fdt blob at 0xc4000000
   Loading Kernel Image
   Loading Device Tree to cffea000, end cffff2c5 ... OK

Starting kernel ...

prefetch abort
pc : [<0c20003c>]          lr : [<d9c47573>]
reloc pc : [<f26ba03c>]    lr : [<c0101573>]
sp : d7aeb278  ip : d7c45102     fp : c2000000
r10: d9c47695  r9 : d7affec0     r8 : d7c5a628
r7 : 00000000  r6 : 0c200040     r5 : d9cf4030  r4 : 00000000
r3 : 000122c6  r2 : cffea000     r1 : 00000000  r0 : 00000000
Flags: nZCv  IRQs off  FIQs off  Mode SVC_32
Code: data abort
pc : [<d9c47920>]          lr : [<d9c4790d>]
reloc pc : [<c0101920>]    lr : [<c010190d>]
sp : d7aeb180  ip : 0000003c     fp : c2000000
r10: d9cd478a  r9 : d7affec0     r8 : d9cd4782
r7 : 00000000  r6 : 00000008     r5 : 0c20003c  r4 : fffffffc
r3 : 00000001  r2 : 00000028     r1 : 00000000  r0 : 00000006
Flags: NzCv  IRQs on  FIQs on  Mode SVC_32 (T)
Code: 2f00 d1e9 2c00 dce9 (f855) 2024
Resetting CPU ...

resetting ...
INFO:    PSCI Power Domain Map:
INFO:      Domain Node : Level 1, parent_node -1, State ON (0x0)
INFO:      Domain Node : Level 0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0x0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0xffffffff, parent_node 0, State OFF (0x2)
NOTICE:  CPU: STM32MP157AAC Rev.Z
NOTICE:  Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
INFO:    Reset reason (0x54):
INFO:      System reset generated by MPU (MPSYSRST)
INFO:    Using EMMC
INFO:      Instance 2
INFO:    Boot used partition fsbl1
NOTICE:  BL2: v2.2-r1.0(debug):85b7ab8
NOTICE:  BL2: Built : 09:15:17, Aug 16 2026
INFO:    Using crypto library 'stm32_crypto_lib'
INFO:    BL2: Doing platform setup
INFO:    RAM: DDR3-1066/888 bin G 2x2Gb 533MHz v1.44
INFO:    Memory size = 0x20000000 (512 MB)
INFO:    BL2 runs SP_MIN setup
INFO:    BL2: Loading image id 4
INFO:    Loading image id=4 at address 0x2ffed000
INFO:    Image id=4 loaded: 0x2ffed000 - 0x2ffff000
INFO:    BL2: Loading image id 5
INFO:    Loading image id=5 at address 0xc0100000
INFO:    STM32 Image size : 869897
INFO:    Image id=5 loaded: 0xc0100000 - 0xc01d4609
WARNING: Skip signature check (header option)
NOTICE:  ROTPK is not deployed on platform. Skipping ROTPK verification.
```

```
NOTICE:  BL2: Booting BL32
INFO:    Entry point address = 0x2ffed000
INFO:    SPSR = 0x1d3
NOTICE:  SP_MIN: v2.2-r1.0(debug):85b7ab8
NOTICE:  SP_MIN: Built : 09:15:17, Aug 16 2026
INFO:    ARM GICv2 driver initialized
INFO:    stm32mp IWDG1 (12): Secure
INFO:    ETZPC: CRYP1 (9) could be non secure
INFO:    SP_MIN: Initializing runtime services
INFO:    SP_MIN: Preparing exit to normal world


U-Boot 2020.01-stm32mp-r1 (Aug 16 2026 - 09:15:35 +0800)

CPU: STM32MP157AAC Rev.Z
Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
Board: stm32mp1 in trusted mode (st,stm32mp157c-100ask-512d-v1)
DRAM:  512 MiB
Clocks:
- MPU : 650 MHz
- MCU : 208.878 MHz
- AXI : 266.500 MHz
- PER : 24 MHz
- DDR : 533 MHz
WDT:   Started with servicing (32s timeout)
NAND:  0 MiB
MMC:   STM32 SD/MMC: 0, STM32 SD/MMC: 1
Loading Environment from MMC... OK
In:    serial
Out:   serial
Err:   serial
Net:   eth0: ethernet@5800a000
Press SPACE to abort autoboot in 1 seconds
STM32MP> mmc dev 1
switch to partitions #0, OK
mmc1(part 0) is current device
STM32MP> run bootcmd_stm32mp
Boot over mmc1!
switch to partitions #0, OK
mmc1(part 0) is current device
Scanning mmc 1:2...
Found /mmc1_extlinux/stm32mp157c-100ask-512d-v1_extlinux.conf
Retrieving file: /mmc1_extlinux/stm32mp157c-100ask-512d-v1_extlinux.conf
210 bytes read in 38 ms (4.9 KiB/s)
1:      100ask-lcd
Retrieving file: /uImage
8171336 bytes read in 316 ms (24.7 MiB/s)
append: root=/dev/mmcblk2p3 rootwait rw console=ttySTM0,115200
Retrieving file: /stm32mp157c-100ask-512d-lcd-v1.dtb
74438 bytes read in 40 ms (1.8 MiB/s)
## Booting kernel from Legacy Image at c2000000 ...
   Image Name:   Linux-5.4.31-g893d15a84
   Created:      2026-10-21  11:38:18 UTC
   Image Type:   ARM Linux Kernel Image (uncompressed)
```

```
   Data Size:    8171272 Bytes = 7.8 MiB
   Load Address: 0c200040
   Entry Point:  0c200040
   Verifying Checksum ... OK
## Flattened Device Tree blob at c4000000
   Booting using the fdt blob at 0xc4000000
   Loading Kernel Image
   Loading Device Tree to cffea000, end cffff2c5 ... OK

Starting kernel ...

prefetch abort
pc : [<0c20003c>]          lr : [<d9c47573>]
reloc pc : [<f26ba03c>]    lr : [<c0101573>]
sp : d7aeb2f8  ip : d7c45102     fp : c2000000
r10: d9c47695  r9 : d7affec0     r8 : d7c56008
r7 : 00000000  r6 : 0c200040     r5 : d9cf4030  r4 : 00000000
r3 : 000122c6  r2 : cffea000     r1 : 00000000  r0 : 00000000
Flags: nZCv  IRQs off  FIQs off  Mode SVC_32
Code: data abort
pc : [<d9c47920>]          lr : [<d9c4790d>]
reloc pc : [<c0101920>]    lr : [<c010190d>]
sp : d7aeb200  ip : 0000003c     fp : c2000000
r10: d9cd478a  r9 : d7affec0     r8 : d9cd4782
r7 : 00000000  r6 : 00000008     r5 : 0c20003c  r4 : fffffffc
r3 : 00000001  r2 : 00000028     r1 : 00000000  r0 : 00000006
Flags: NzCv  IRQs on  FIQs on  Mode SVC_32 (T)
Code: 2f00 d1e9 2c00 dce9 (f855) 2024
Resetting CPU ...

resetting ...
INFO:    PSCI Power Domain Map:
INFO:      Domain Node : Level 1, parent_node -1, State ON (0x0)
INFO:      Domain Node : Level 0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0x0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0xffffffff, parent_node 0, State OFF (0x2)
NOTICE:  CPU: STM32MP157AAC Rev.Z
NOTICE:  Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
INFO:    Reset reason (0x54):
INFO:      System reset generated by MPU (MPSYSRST)
INFO:    Using EMMC
INFO:      Instance 2
INFO:    Boot used partition fsbl1
NOTICE:  BL2: v2.2-r1.0(debug):85b7ab8
NOTICE:  BL2: Built : 09:15:17, Aug 16 2026
INFO:    Using crypto library 'stm32_crypto_lib'
INFO:    BL2: Doing platform setup
INFO:    RAM: DDR3-1066/888 bin G 2x2Gb 533MHz v1.44
INFO:    Memory size = 0x20000000 (512 MB)
INFO:    BL2 runs SP_MIN setup
INFO:    BL2: Loading image id 4
INFO:    Loading image id=4 at address 0x2ffed000
INFO:    Image id=4 loaded: 0x2ffed000 - 0x2ffff000
INFO:    BL2: Loading image id 5
```

```
INFO:    Loading image id=5 at address 0xc0100000
INFO:    STM32 Image size : 869897
INFO:    Image id=5 loaded: 0xc0100000 - 0xc01d4609
WARNING: Skip signature check (header option)
NOTICE:  ROTPK is not deployed on platform. Skipping ROTPK verification.
NOTICE:  BL2: Booting BL32
INFO:    Entry point address = 0x2ffed000
INFO:    SPSR = 0x1d3
NOTICE:  SP_MIN: v2.2-r1.0(debug):85b7ab8
NOTICE:  SP_MIN: Built : 09:15:17, Aug 16 2026
INFO:    ARM GICv2 driver initialized
INFO:    stm32mp IWDG1 (12): Secure
INFO:    ETZPC: CRYP1 (9) could be non secure
INFO:    SP_MIN: Initializing runtime services
INFO:    SP_MIN: Preparing exit to normal world


U-Boot 2020.01-stm32mp-r1 (Aug 16 2026 - 09:15:35 +0800)

CPU: STM32MP157AAC Rev.Z
Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
Board: stm32mp1 in trusted mode (st,stm32mp157c-100ask-512d-v1)
DRAM:  512 MiB
Clocks:
- MPU : 650 MHz
- MCU : 208.878 MHz
- AXI : 266.500 MHz
- PER : 24 MHz
- DDR : 533 MHz
WDT:   Started with servicing (32s timeout)
NAND:  0 MiB
MMC:   STM32 SD/MMC: 0, STM32 SD/MMC: 1
Loading Environment from MMC... OK
In:    serial
Out:   serial
Err:   serial
Net:   eth0: ethernet@5800a000
Press SPACE to abort autoboot in 1 seconds
Boot over mmc1!
switch to partitions #0, OK
mmc1(part 0) is current device
Scanning mmc 1:2...
Found /mmc1_extlinux/stm32mp157c-100ask-512d-v1_extlinux.conf
Retrieving file: /mmc1_extlinux/stm32mp157c-100ask-512d-v1_extlinux.conf
210 bytes read in 38 ms (4.9 KiB/s)
1:      100ask-lcd
Retrieving file: /uImage
8171336 bytes read in 316 ms (24.7 MiB/s)
append: root=/dev/mmcblk2p3 rootwait rw console=ttySTM0,115200
Retrieving file: /stm32mp157c-100ask-512d-lcd-v1.dtb
74438 bytes read in 40 ms (1.8 MiB/s)
## Booting kernel from Legacy Image at c2000000 ...
   Image Name:   Linux-5.4.31-g893d15a84
   Created:      2026-10-21  11:38:18 UTC
```

```
   Image Type:   ARM Linux Kernel Image (uncompressed)
   Data Size:    8171272 Bytes = 7.8 MiB
   Load Address: 0c200040
   Entry Point:  0c200040
   Verifying Checksum ... OK
## Flattened Device Tree blob at c4000000
   Booting using the fdt blob at 0xc4000000
   Loading Kernel Image
   Loading Device Tree to cffea000, end cffff2c5 ... OK

Starting kernel ...

prefetch abort
pc : [<0c20003c>]          lr : [<d9c47573>]
reloc pc : [<f26ba03c>]    lr : [<c0101573>]
sp : d7aeb278  ip : d7c45102     fp : c2000000
r10: d9c47695  r9 : d7affec0     r8 : d7c5a628
r7 : 00000000  r6 : 0c200040     r5 : d9cf4030  r4 : 00000000
r3 : 000122c6  r2 : cffea000     r1 : 00000000  r0 : 00000000
Flags: nZCv  IRQs off  FIQs off  Mode SVC_32
Code: data abort
pc : [<d9c47920>]          lr : [<d9c4790d>]
reloc pc : [<c0101920>]    lr : [<c010190d>]
sp : d7aeb180  ip : 0000003c     fp : c2000000
r10: d9cd478a  r9 : d7affec0     r8 : d9cd4782
r7 : 00000000  r6 : 00000008     r5 : 0c20003c  r4 : fffffffc
r3 : 00000001  r2 : 00000028     r1 : 00000000  r0 : 00000006
Flags: NzCv  IRQs on  FIQs on  Mode SVC_32 (T)
Code: 2f00 d1e9 2c00 dce9 (f855) 2024
Resetting CPU ...

resetting ...
INFO:    PSCI Power Domain Map:
INFO:      Domain Node : Level 1, parent_node -1, State ON (0x0)
INFO:      Domain Node : Level 0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0x0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0xffffffff, parent_node 0, State OFF (0x2)
NOTICE:  CPU: STM32MP157AAC Rev.Z
NOTICE:  Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
INFO:    Reset reason (0x54):
INFO:      System reset generated by MPU (MPSYSRST)
INFO:    Using EMMC
INFO:      Instance 2
INFO:    Boot used partition fsbl1
NOTICE:  BL2: v2.2-r1.0(debug):85b7ab8
NOTICE:  BL2: Built : 09:15:17, Aug 16 2026
INFO:    Using crypto library 'stm32_crypto_lib'
INFO:    BL2: Doing platform setup
INFO:    RAM: DDR3-1066/888 bin G 2x2Gb 533MHz v1.44
INFO:    Memory size = 0x20000000 (512 MB)
INFO:    BL2 runs SP_MIN setup
INFO:    BL2: Loading image id 4
INFO:    Loading image id=4 at address 0x2ffed000
INFO:    Image id=4 loaded: 0x2ffed000 - 0x2ffff000
```

```
INFO:    BL2: Loading image id 5
INFO:    Loading image id=5 at address 0xc0100000
INFO:    STM32 Image size : 869897
INFO:    Image id=5 loaded: 0xc0100000 - 0xc01d4609
WARNING: Skip signature check (header option)
NOTICE:  ROTPK is not deployed on platform. Skipping ROTPK verification.
NOTICE:  BL2: Booting BL32
INFO:    Entry point address = 0x2ffed000
INFO:    SPSR = 0x1d3
NOTICE:  SP_MIN: v2.2-r1.0(debug):85b7ab8
NOTICE:  SP_MIN: Built : 09:15:17, Aug 16 2026
INFO:    ARM GICv2 driver initialized
INFO:    stm32mp IWDG1 (12): Secure
INFO:    ETZPC: CRYP1 (9) could be non secure
INFO:    SP_MIN: Initializing runtime services
INFO:    SP_MIN: Preparing exit to normal world


U-Boot 2020.01-stm32mp-r1 (Aug 16 2026 - 09:15:35 +0800)

CPU: STM32MP157AAC Rev.Z
Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
Board: stm32mp1 in trusted mode (st,stm32mp157c-100ask-512d-v1)
DRAM:  512 MiB
Clocks:
- MPU : 650 MHz
- MCU : 208.878 MHz
- AXI : 266.500 MHz
- PER : 24 MHz
- DDR : 533 MHz
WDT:   Started with servicing (32s timeout)
NAND:  0 MiB
MMC:   STM32 SD/MMC: 0, STM32 SD/MMC: 1
Loading Environment from MMC... OK
In:    serial
Out:   serial
Err:   serial
Net:   eth0: ethernet@5800a000
Press SPACE to abort autoboot in 1 seconds
STM32MP> setenv ipaddr 192.168.145.129
STM32MP> setenv serverip 192.168.145.128
STM32MP> saveenv
Saving Environment to MMC... Writing to redundant MMC(1)... OK
STM32MP> tftp 0xc4200000 uImage
yt8511_config_out_125m, phy clk out, vall=8057
ethernet@5800a000 Waiting for PHY auto negotiation to complete... done
Using ethernet@5800a000 device
TFTP from server 192.168.145.128; our IP address is 192.168.145.129
Filename 'uImage'.
Load address: 0xc4200000
Loading: *
TFTP error: 'File not found' (1)
Not retrying...
STM32MP> tftp 0xc4200000 uImage
```

```
Using ethernet@5800a000 device
TFTP from server 192.168.145.128; our IP address is 192.168.145.129
Filename 'uImage'.
Load address: 0xc4200000
Loading: #################################################################
         #################################################################
         #################################################################
         ##############################################T ##################
         #################################################################
         #################################################################
         #################################################################
         #################################################################
         ####################################
         472.7 KiB/s
done
Bytes transferred = 8187672 (7cef18 hex)
STM32MP> tftp 0xc4000000 stm32mp157c-100ask-512d-v1.dtb
Using ethernet@5800a000 device
TFTP from server 192.168.145.128; our IP address is 192.168.145.129
Filename 'stm32mp157c-100ask-512d-v1.dtb'.
Load address: 0xc4000000
Loading: ########
         631.8 KiB/s
done
Bytes transferred = 107530 (1a40a hex)
STM32MP> mmc dev 1
switch to partitions #0, OK
mmc1(part 0) is current device
STM32MP> fatload mmc 1:2 0xc4000000 uImage
8171336 bytes read in 316 ms (24.7 MiB/s)
STM32MP> fatload mmc 1:2 0xc4200000 uImage
8171336 bytes read in 315 ms (24.7 MiB/s)
STM32MP> fatload mmc 1:2 0xc4000000 stm32mp157c-100ask-512d-v1.dtb
73271 bytes read in 41 ms (1.7 MiB/s)
STM32MP> setenv bootargs 'root=/dev/mmcblk1p3 rootwait rw console=ttySTM0,115
200'
STM32MP> bootm 0xc4200000 - 0xc4000000
## Booting kernel from Legacy Image at c4200000 ...
   Image Name:   Linux-5.4.31-g893d15a84
   Created:      2026-10-21  11:38:18 UTC
   Image Type:   ARM Linux Kernel Image (uncompressed)
   Data Size:    8171272 Bytes = 7.8 MiB
   Load Address: 0c200040
   Entry Point:  0c200040
   Verifying Checksum ... OK
## Flattened Device Tree blob at c4000000
   Booting using the fdt blob at 0xc4000000
   Loading Kernel Image
   Loading Device Tree to cffeb000, end cffffe36 ... OK

Starting kernel ...

prefetch abort
pc : [<0c20003c>]          lr : [<d9c47573>]
```

```
reloc pc : [<f26ba03c>]    lr : [<c0101573>]
sp : d7aec9a0  ip : d7c45102     fp : c4200000
r10: d9c47695  r9 : d7affec0     r8 : d7c4b4c0
r7 : 00000000  r6 : 0c200040     r5 : d9cf4030  r4 : 00000000
r3 : 00011e37  r2 : cffeb000     r1 : 00000000  r0 : 00000000
Flags: nZCv  IRQs off  FIQs off  Mode SVC_32
Code: data abort
pc : [<d9c47920>]          lr : [<d9c4790d>]
reloc pc : [<c0101920>]    lr : [<c010190d>]
sp : d7aec8a8  ip : 0000003c     fp : c4200000
r10: d9cd478a  r9 : d7affec0     r8 : d9cd4782
r7 : 00000000  r6 : 00000008     r5 : 0c20003c  r4 : fffffffc
r3 : 00000001  r2 : 00000028     r1 : 00000000  r0 : 00000006
Flags: NzCv  IRQs on  FIQs on  Mode SVC_32 (T)
Code: 2f00 d1e9 2c00 dce9 (f855) 2024
Resetting CPU ...

resetting ...
INFO:    PSCI Power Domain Map:
INFO:      Domain Node : Level 1, parent_node -1, State ON (0x0)
INFO:      Domain Node : Level 0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0x0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0xffffffff, parent_node 0, State OFF (0x2)
NOTICE:  CPU: STM32MP157AAC Rev.Z
NOTICE:  Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
INFO:    Reset reason (0x54):
INFO:      System reset generated by MPU (MPSYSRST)
INFO:    Using EMMC
INFO:      Instance 2
INFO:    Boot used partition fsbl1
NOTICE:  BL2: v2.2-r1.0(debug):85b7ab8
NOTICE:  BL2: Built : 09:15:17, Aug 16 2026
INFO:    Using crypto library 'stm32_crypto_lib'
INFO:    BL2: Doing platform setup
INFO:    RAM: DDR3-1066/888 bin G 2x2Gb 533MHz v1.44
INFO:    Memory size = 0x20000000 (512 MB)
INFO:    BL2 runs SP_MIN setup
INFO:    BL2: Loading image id 4
INFO:    Loading image id=4 at address 0x2ffed000
INFO:    Image id=4 loaded: 0x2ffed000 - 0x2ffff000
INFO:    BL2: Loading image id 5
INFO:    Loading image id=5 at address 0xc0100000
INFO:    STM32 Image size : 869897
INFO:    Image id=5 loaded: 0xc0100000 - 0xc01d4609
WARNING: Skip signature check (header option)
NOTICE:  ROTPK is not deployed on platform. Skipping ROTPK verification.
NOTICE:  BL2: Booting BL32
INFO:    Entry point address = 0x2ffed000
INFO:    SPSR = 0x1d3
NOTICE:  SP_MIN: v2.2-r1.0(debug):85b7ab8
NOTICE:  SP_MIN: Built : 09:15:17, Aug 16 2026
INFO:    ARM GICv2 driver initialized
INFO:    stm32mp IWDG1 (12): Secure
INFO:    ETZPC: CRYP1 (9) could be non secure
```

```
INFO:    SP_MIN: Initializing runtime services
INFO:    SP_MIN: Preparing exit to normal world


U-Boot 2020.01-stm32mp-r1 (Aug 16 2026 - 09:15:35 +0800)

CPU: STM32MP157AAC Rev.Z
Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
Board: stm32mp1 in trusted mode (st,stm32mp157c-100ask-512d-v1)
DRAM:  512 MiB
Clocks:
- MPU : 650 MHz
- MCU : 208.878 MHz
- AXI : 266.500 MHz
- PER : 24 MHz
- DDR : 533 MHz
WDT:   Started with servicing (32s timeout)
NAND:  0 MiB
MMC:   STM32 SD/MMC: 0, STM32 SD/MMC: 1
Loading Environment from MMC... OK
In:    serial
Out:   serial
Err:   serial
Net:   eth0: ethernet@5800a000
Press SPACE to abort autoboot in 1 seconds
Boot over mmc1!
switch to partitions #0, OK
mmc1(part 0) is current device
Scanning mmc 1:2...
Found /mmc1_extlinux/stm32mp157c-100ask-512d-v1_extlinux.conf
Retrieving file: /mmc1_extlinux/stm32mp157c-100ask-512d-v1_extlinux.conf
210 bytes read in 37 ms (4.9 KiB/s)
1:      100ask-lcd
Retrieving file: /uImage
8171336 bytes read in 315 ms (24.7 MiB/s)
append: root=/dev/mmcblk2p3 rootwait rw console=ttySTM0,115200
Retrieving file: /stm32mp157c-100ask-512d-lcd-v1.dtb
74438 bytes read in 40 ms (1.8 MiB/s)
## Booting kernel from Legacy Image at c2000000 ...
   Image Name:   Linux-5.4.31-g893d15a84
   Created:      2026-10-21  11:38:18 UTC
   Image Type:   ARM Linux Kernel Image (uncompressed)
   Data Size:    8171272 Bytes = 7.8 MiB
   Load Address: 0c200040
   Entry Point:  0c200040
   Verifying Checksum ... OK
## Flattened Device Tree blob at c4000000
   Booting using the fdt blob at 0xc4000000
   Loading Kernel Image
   Loading Device Tree to cffea000, end cffff2c5 ... OK

Starting kernel ...

prefetch abort
```

```
pc : [<0c20003c>]          lr : [<d9c47573>]
reloc pc : [<f26ba03c>]    lr : [<c0101573>]
sp : d7aeb278  ip : d7c45102     fp : c2000000
r10: d9c47695  r9 : d7affec0     r8 : d7c5a1c8
r7 : 00000000  r6 : 0c200040     r5 : d9cf4030  r4 : 00000000
r3 : 000122c6  r2 : cffea000     r1 : 00000000  r0 : 00000000
Flags: nZCv  IRQs off  FIQs off  Mode SVC_32
Code: data abort
pc : [<d9c47920>]          lr : [<d9c4790d>]
reloc pc : [<c0101920>]    lr : [<c010190d>]
sp : d7aeb180  ip : 0000003c     fp : c2000000
r10: d9cd478a  r9 : d7affec0     r8 : d9cd4782
r7 : 00000000  r6 : 00000008     r5 : 0c20003c  r4 : fffffffc
r3 : 00000001  r2 : 00000028     r1 : 00000000  r0 : 00000006
Flags: NzCv  IRQs on  FIQs on  Mode SVC_32 (T)
Code: 2f00 d1e9 2c00 dce9 (f855) 2024
Resetting CPU ...

resetting ...
INFO:    PSCI Power Domain Map:
INFO:      Domain Node : Level 1, parent_node -1, State ON (0x0)
INFO:      Domain Node : Level 0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0x0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0xffffffff, parent_node 0, State OFF (0x2)
NOTICE:  CPU: STM32MP157AAC Rev.Z
NOTICE:  Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
INFO:    Reset reason (0x54):
INFO:      System reset generated by MPU (MPSYSRST)
INFO:    Using EMMC
INFO:      Instance 2
INFO:    Boot used partition fsbl1
NOTICE:  BL2: v2.2-r1.0(debug):85b7ab8
NOTICE:  BL2: Built : 09:15:17, Aug 16 2026
INFO:    Using crypto library 'stm32_crypto_lib'
INFO:    BL2: Doing platform setup
INFO:    RAM: DDR3-1066/888 bin G 2x2Gb 533MHz v1.44
INFO:    Memory size = 0x20000000 (512 MB)
INFO:    BL2 runs SP_MIN setup
INFO:    BL2: Loading image id 4
INFO:    Loading image id=4 at address 0x2ffed000
INFO:    Image id=4 loaded: 0x2ffed000 - 0x2ffff000
INFO:    BL2: Loading image id 5
INFO:    Loading image id=5 at address 0xc0100000
INFO:    STM32 Image size : 869897
INFO:    Image id=5 loaded: 0xc0100000 - 0xc01d4609
WARNING: Skip signature check (header option)
NOTICE:  ROTPK is not deployed on platform. Skipping ROTPK verification.
NOTICE:  BL2: Booting BL32
INFO:    Entry point address = 0x2ffed000
INFO:    SPSR = 0x1d3
NOTICE:  SP_MIN: v2.2-r1.0(debug):85b7ab8
NOTICE:  SP_MIN: Built : 09:15:17, Aug 16 2026
INFO:    ARM GICv2 driver initialized
INFO:    stm32mp IWDG1 (12): Secure
```

```
INFO:    ETZPC: CRYP1 (9) could be non secure
INFO:    SP_MIN: Initializing runtime services
INFO:    SP_MIN: Preparing exit to normal world


U-Boot 2020.01-stm32mp-r1 (Aug 16 2026 - 09:15:35 +0800)

CPU: STM32MP157AAC Rev.Z
Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
Board: stm32mp1 in trusted mode (st,stm32mp157c-100ask-512d-v1)
DRAM:  512 MiB
Clocks:
- MPU : 650 MHz
- MCU : 208.878 MHz
- AXI : 266.500 MHz
- PER : 24 MHz
- DDR : 533 MHz
WDT:   Started with servicing (32s timeout)
NAND:  0 MiB
MMC:   STM32 SD/MMC: 0, STM32 SD/MMC: 1
Loading Environment from MMC... OK
In:    serial
Out:   serial
Err:   serial
Net:   eth0: ethernet@5800a000
Press SPACE to abort autoboot in 1 seconds
STM32MP> tftp 0xc4200000 uImage
yt8511_config_out_125m, phy clk out, vall=8057
ethernet@5800a000 Waiting for PHY auto negotiation to complete... done
Using ethernet@5800a000 device
TFTP from server 192.168.145.128; our IP address is 192.168.145.129
Filename 'uImage'.
Load address: 0xc4200000
Loading: ##################################################################
         ##################################################################
         ##################################################################
         ##################################################T ###############
         ##################################################################
         ##################################################################
         ##################################################################
         ##################################################################
         ######################################
         364.3 KiB/s
done
Bytes transferred = 8187672 (7cef18 hex)
STM32MP> tftp 0xc4000000 stm32mp157c-100ask-512d-v1.dtb
Using ethernet@5800a000 device
TFTP from server 192.168.145.128; our IP address is 192.168.145.129
Filename 'stm32mp157c-100ask-512d-v1.dtb'.
Load address: 0xc4000000
Loading: ########
         643.6 KiB/s
done
Bytes transferred = 107530 (1a40a hex)
```

```
STM32MP> fatwrite mmc 1:2 0xc4200000 uImage ${filesize}
Unknown command 'fatwrite' - try 'help'
STM32MP> setenv bootargs 'root=/dev/mmcblk1p3 rootwait rw console=ttySTM0,115
200'
STM32MP> bootm 0xc4200000 - 0xc4000000
## Booting kernel from Legacy Image at c4200000 ...
   Image Name:   Linux-5.4.31
   Created:      2022-10-27  11:09:36 UTC
   Image Type:   ARM Linux Kernel Image (uncompressed)
   Data Size:    8187608 Bytes = 7.8 MiB
   Load Address: c2000040
   Entry Point:  c2000040
   Verifying Checksum ... OK
## Flattened Device Tree blob at c4000000
   Booting using the fdt blob at 0xc4000000
   Loading Kernel Image
   Loading Device Tree to cffe2000, end cffff409 ... OK

Starting kernel ...

[    0.000000] Booting Linux on physical CPU 0x0
[    0.000000] Linux version 5.4.31 (book@virtual-machine) (gcc version 8.4.0
 (Buildroot 2020.02-g8ad4a8c-dirty)) #1 SMP PREEMPT Thu Oct 27 11:04:26 UTC 2
022
[    0.000000] CPU: ARMv7 Processor [410fc075] revision 5 (ARMv7), cr=10c5387
d
[    0.000000] CPU: div instructions available: patching division code
[    0.000000] CPU: PIPT / VIPT nonaliasing data cache, VIPT aliasing instruc
tion cache
[    0.000000] OF: fdt: Machine model: 100ASK YA157C v2 www.100ask.com
[    0.000000] Memory policy: Data cache writealloc
[    0.000000] Reserved memory: created DMA memory pool at 0x10000000, size 0
 MiB
[    0.000000] OF: reserved mem: initialized node mcuram2@10000000, compatibl
e id shared-dma-pool
[    0.000000] Reserved memory: created DMA memory pool at 0x10040000, size 0
 MiB
[    0.000000] OF: reserved mem: initialized node vdev0vring0@10040000, compa
tible id shared-dma-pool
[    0.000000] Reserved memory: created DMA memory pool at 0x10041000, size 0
 MiB
[    0.000000] OF: reserved mem: initialized node vdev0vring1@10041000, compa
tible id shared-dma-pool
[    0.000000] Reserved memory: created DMA memory pool at 0x10042000, size 0
 MiB
[    0.000000] OF: reserved mem: initialized node vdev0buffer@10042000, compa
tible id shared-dma-pool
[    0.000000] Reserved memory: created DMA memory pool at 0x30000000, size 0
 MiB
[    0.000000] OF: reserved mem: initialized node mcuram@30000000, compatible
 id shared-dma-pool
[    0.000000] Reserved memory: created DMA memory pool at 0x38000000, size 0
 MiB
[    0.000000] OF: reserved mem: initialized node retram@38000000, compatible
```

```
 id shared-dma-pool
[    0.000000] cma: Reserved 128 MiB at 0xd2000000
[    0.000000] psci: probing for conduit method from DT.
[    0.000000] psci: PSCIv1.1 detected in firmware.
[    0.000000] psci: Using standard PSCI v0.2 function IDs
[    0.000000] psci: MIGRATE_INFO_TYPE not supported.
[    0.000000] psci: SMC Calling Convention v1.0
[    0.000000] percpu: Embedded 20 pages/cpu s50124 r8192 d23604 u81920
[    0.000000] Built 1 zonelists, mobility grouping on.  Total pages: 113536
[    0.000000] Kernel command line: root=/dev/mmcblk1p3 rootwait rw console=t
tySTM0,115200
[    0.000000] Dentry cache hash table entries: 65536 (order: 6, 262144 bytes
, linear)
[    0.000000] Inode-cache hash table entries: 32768 (order: 5, 131072 bytes,
 linear)
[    0.000000] mem auto-init: stack:off, heap alloc:off, heap free:off
[    0.000000] Memory: 303816K/458752K available (12288K kernel code, 962K rw
data, 3516K rodata, 1024K init, 226K bss, 23864K reserved, 131072K cma-reserv
ed, 0K highmem)
[    0.000000] SLUB: HWalign=64, Order=0-3, MinObjects=0, CPUs=2, Nodes=1
[    0.000000] rcu: Preemptible hierarchical RCU implementation.
[    0.000000] rcu:     RCU event tracing is enabled.
[    0.000000]  Tasks RCU enabled.
[    0.000000] rcu: RCU calculated value of scheduler-enlistment delay is 10
jiffies.
[    0.000000] NR_IRQS: 16, nr_irqs: 16, preallocated irqs: 16
[    0.000000] random: get_random_bytes called from start_kernel+0x320/0x4f4
with crng_init=0
[    0.000000] arch_timer: cp15 timer(s) running at 24.00MHz (virt).
[    0.000000] clocksource: arch_sys_counter: mask: 0xffffffffffffff max_cycl
es: 0x588fe9dc0, max_idle_ns: 440795202592 ns
[    0.000009] sched_clock: 56 bits at 24MHz, resolution 41ns, wraps every 43
98046511097ns
[    0.000027] Switching to timer-based delay loop, resolution 41ns
[    0.001997] Console: colour dummy device 80x30
[    0.002049] Calibrating delay loop (skipped), value calculated using timer
 frequency.. 48.00 BogoMIPS (lpj=240000)
[    0.002070] pid_max: default: 32768 minimum: 301
[    0.002333] Mount-cache hash table entries: 1024 (order: 0, 4096 bytes, li
near)
[    0.002356] Mountpoint-cache hash table entries: 1024 (order: 0, 4096 byte
s, linear)
[    0.003654] CPU: Testing write buffer coherency: ok
[    0.004185] /cpus/cpu@0 missing clock-frequency property
[    0.004220] /cpus/cpu@1 missing clock-frequency property
[    0.004239] CPU0: thread -1, cpu 0, socket 0, mpidr 80000000
[    0.059762] Setting up static identity map for 0xc0100000 - 0xc0100060
[    0.079737] rcu: Hierarchical SRCU implementation.
[    0.105116] smp: Bringing up secondary CPUs ...
[    0.180379] CPU1: thread -1, cpu 1, socket 0, mpidr 80000001
[    0.180597] smp: Brought up 1 node, 2 CPUs
[    0.180622] SMP: Total of 2 processors activated (96.00 BogoMIPS).
[    0.180633] CPU: All CPU(s) started in SVC mode.
[    0.181479] devtmpfs: initialized
```

```
[    0.214442] VFP support v0.3: implementor 41 architecture 2 part 30 varian
t 7 rev 5
[    0.215070] clocksource: jiffies: mask: 0xffffffff max_cycles: 0xffffffff,
 max_idle_ns: 19112604462750000 ns
[    0.215105] futex hash table entries: 512 (order: 3, 32768 bytes, linear)
[    0.221970] pinctrl core: initialized pinctrl subsystem
[    0.224387] NET: Registered protocol family 16
[    0.233136] DMA: preallocated 256 KiB pool for atomic coherent allocations
[    0.235730] hw-breakpoint: found 5 (+1 reserved) breakpoint and 4 watchpoi
nt registers.
[    0.235752] hw-breakpoint: maximum watchpoint size is 8 bytes.
[    0.236182] Serial: AMBA PL011 UART driver
[    0.239412] stm32-pm-domain pm_domain: domain core-ret-power-domain regist
ered
[    0.239471] stm32-pm-domain pm_domain: subdomain core-power-domain registe
red
[    0.239488] stm32-pm-domain pm_domain: domains probed
[    0.291175] gpio-regulator regulators: No 'states' property found
[    0.291220] gpio-regulator: probe of regulators failed with error -22
[    0.292675] SCSI subsystem initialized
[    0.293531] usbcore: registered new interface driver usbfs
[    0.293617] usbcore: registered new interface driver hub
[    0.293776] usbcore: registered new device driver usb
[    0.294165] pps_core: LinuxPPS API ver. 1 registered
[    0.294180] pps_core: Software ver. 5.3.6 - Copyright 2005-2007 Rodolfo Gi
ometti <giometti@linux.it>
[    0.294213] PTP clock support registered
[    0.294570] EDAC MC: Ver: 3.0.0
[    0.295877] Advanced Linux Sound Architecture Driver Initialized.
[    0.296746] Bluetooth: Core ver 2.22
[    0.296854] NET: Registered protocol family 31
[    0.296868] Bluetooth: HCI device and connection manager initialized
[    0.296891] Bluetooth: HCI socket layer initialized
[    0.296908] Bluetooth: L2CAP socket layer initialized
[    0.296945] Bluetooth: SCO socket layer initialized
[    0.297937] clocksource: Switched to clocksource arch_sys_counter
[    1.057856] thermal_sys: Registered thermal governor 'step_wise'
[    1.058565] NET: Registered protocol family 2
[    1.059687] tcp_listen_portaddr_hash hash table entries: 512 (order: 0, 61
44 bytes, linear)
[    1.059740] TCP established hash table entries: 4096 (order: 2, 16384 byte
s, linear)
[    1.059808] TCP bind hash table entries: 4096 (order: 3, 32768 bytes, line
ar)
[    1.059905] TCP: Hash tables configured (established 4096 bind 4096)
[    1.060050] UDP hash table entries: 256 (order: 1, 8192 bytes, linear)
[    1.060098] UDP-Lite hash table entries: 256 (order: 1, 8192 bytes, linear
)
[    1.060417] NET: Registered protocol family 1
[    1.061406] RPC: Registered named UNIX socket transport module.
[    1.061426] RPC: Registered udp transport module.
[    1.061437] RPC: Registered tcp transport module.
[    1.061447] RPC: Registered tcp NFSv4.1 backchannel transport module.
[    1.063321] hw perfevents: enabled with armv7_cortex_a7 PMU driver, 5 coun
```

```
ters available
[    1.065419] Initialise system trusted keyrings
[    1.065802] workingset: timestamp_bits=14 max_order=17 bucket_order=3
[    1.078504] squashfs: version 4.0 (2009/01/31) Phillip Lougher
[    1.089902] NFS: Registering the id_resolver key type
[    1.089954] Key type id_resolver registered
[    1.089967] Key type id_legacy registered
[    1.090227] ntfs: driver 2.1.32 [Flags: R/O].
[    1.090730] jffs2: version 2.2. (NAND) © 2001-2006 Red Hat, Inc.
[    1.091525] fuse: init (API version 7.31)
[    1.146137] NET: Registered protocol family 38
[    1.146166] Key type asymmetric registered
[    1.146179] Asymmetric key parser 'x509' registered
[    1.146275] Block layer SCSI generic (bsg) driver version 0.4 loaded (majo
r 246)
[    1.146292] io scheduler mq-deadline registered
[    1.146304] io scheduler kyber registered
[    1.156577] STM32 USART driver initialized
[    1.158299] ar8035 phy init
[    1.158780] the phy gpio failed
[    1.179577] brd: module loaded
[    1.194004] loop: module loaded
[    1.200279] libphy: Fixed MDIO Bus: probed
[    1.202172] CAN device driver interface
[    1.204643] PPP generic driver version 2.4.2
[    1.204978] PPP BSD Compression module registered
[    1.204995] PPP Deflate Compression module registered
[    1.205023] PPP MPPE Compression module registered
[    1.205036] NET: Registered protocol family 24
[    1.205081] SLIP: version 0.8.4-NET3.019-NEWTTY (dynamic channels, max=256
) (6 bit encapsulation enabled).
[    1.205092] CSLIP: code copyright 1989 Regents of the University of Califo
rnia.
[    1.205100] SLIP linefill/keepalive option.
[    1.205115] pegasus: v0.9.3 (2013/04/25), Pegasus/Pegasus II USB Ethernet
driver
[    1.205219] usbcore: registered new interface driver pegasus
[    1.205302] usbcore: registered new interface driver r8152
[    1.205392] usbcore: registered new interface driver asix
[    1.205466] usbcore: registered new interface driver ax88179_178a
[    1.205526] usbcore: registered new interface driver cdc_ether
[    1.205604] usbcore: registered new interface driver smsc75xx
[    1.205682] usbcore: registered new interface driver smsc95xx
[    1.205742] usbcore: registered new interface driver net1080
[    1.205800] usbcore: registered new interface driver cdc_subset
[    1.205868] usbcore: registered new interface driver zaurus
[    1.205958] usbcore: registered new interface driver cdc_ncm
[    1.207599] ehci_hcd: USB 2.0 'Enhanced' Host Controller (EHCI) Driver
[    1.207656] ehci-platform: EHCI generic platform driver
[    1.208326] ohci_hcd: USB 1.1 'Open' Host Controller (OHCI) Driver
[    1.208371] ohci-platform: OHCI generic platform driver
[    1.209394] usbcore: registered new interface driver cdc_acm
[    1.209411] cdc_acm: USB Abstract Control Model driver for USB modems and
ISDN adapters
```

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[    1.209497] usbcore: registered new interface driver usblp
[    1.209560] usbcore: registered new interface driver cdc_wdm
[    1.209620] usbcore: registered new interface driver uas
[    1.209702] usbcore: registered new interface driver usb-storage
[    1.209856] usbcore: registered new interface driver usbserial_generic
[    1.209907] usbserial: USB Serial support registered for generic
[    1.209975] usbcore: registered new interface driver option
[    1.210027] usbserial: USB Serial support registered for GSM modem (1-port
)
[    1.215021] i2c /dev entries driver
[    1.220089] stm32-cpufreq stm32-cpufreq: Failed to get chip info: -517
[    1.224649] sdhci: Secure Digital Host Controller Interface driver
[    1.224665] sdhci: Copyright(c) Pierre Ossman
[    1.224675] Synopsys Designware Multimedia Card Interface Driver
[    1.225253] sdhci-pltfm: SDHCI platform and OF driver helper
[    1.226545] ledtrig-cpu: registered to indicate activity on CPUs
[    1.228954] usbcore: registered new interface driver usbhid
[    1.228974] usbhid: USB HID core driver
[    1.228985] dtbocfg_module_init
[    1.229053] dtbocfg_module_init: OK
[    1.229422] arm-smc-mbox mailbox-0: ARM SMC mailbox enabled.
[    1.229556] arm-smc-mbox mailbox-1: ARM SMC mailbox enabled.
[    1.231338] remoteproc remoteproc0: releasing m4
[    1.234470] Unable to get STM32 DDR PMU clock
[    1.239931] drop_monitor: Initializing network drop monitor service
[    1.240290] NET: Registered protocol family 17
[    1.240323] can: controller area network core (rev 20170425 abi 9)
[    1.240445] NET: Registered protocol family 29
[    1.240462] can: raw protocol (rev 20170425)
[    1.240474] can: broadcast manager protocol (rev 20170425 t)
[    1.240493] can: netlink gateway (rev 20190810) max_hops=1
[    1.240970] Bluetooth: RFCOMM TTY layer initialized
[    1.241015] Bluetooth: RFCOMM socket layer initialized
[    1.241067] Bluetooth: RFCOMM ver 1.11
[    1.241094] Bluetooth: BNEP (Ethernet Emulation) ver 1.3
[    1.241106] Bluetooth: BNEP filters: protocol multicast
[    1.241126] Bluetooth: BNEP socket layer initialized
[    1.241139] Bluetooth: HIDP (Human Interface Emulation) ver 1.2
[    1.241157] Bluetooth: HIDP socket layer initialized
[    1.241739] Key type dns_resolver registered
[    1.241933] ThumbEE CPU extension supported.
[    1.241963] Registering SWP/SWPB emulation handler
[    1.242413] registered taskstats version 1
[    1.242428] Loading compiled-in X.509 certificates
[    1.253692] Loaded X.509 cert 'Build time autogenerated kernel key: a551a3
bcaf9a62271059adf2e1003b2f3df06de0'
[    1.278233] arm-scmi firmware:scmi-0: SCMI Protocol v2.0 'ST:' Firmware ve
rsion 0x0
[    1.282638] remoteproc remoteproc0: releasing m4
[    1.284094] Unable to get STM32 DDR PMU clock
[    1.305233] debugfs: Directory 'cpu0' with parent 'opp' already present!
[    1.315068] remoteproc remoteproc0: releasing m4
[    1.316531] Unable to get STM32 DDR PMU clock
[    1.355589] stm32-mdma 58000000.dma: STM32 MDMA driver registered
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[    1.360168] stm32-dma 48000000.dma: STM32 DMA driver registered
[    1.362316] stm32-dma 48001000.dma: STM32 DMA driver registered
[    1.362964] reg11: supplied by regulator-dummy
[    1.363440] reg18: supplied by regulator-dummy
[    1.363868] usb33: supplied by regulator-dummy
[    1.388350] stm_thermal 50028000.thermal: stm_thermal_probe: Driver initia
lized successfully
[    1.388672] random: fast init done
[    1.391187] debugfs: Directory 'cpu0' with parent 'opp' already present!
[    1.393096] random: crng init done
[    1.395419] stm32-crc32 58009000.crc: Initialized
[    1.397002] stm32-hash 54002000.hash: will run requests pump with realtime
 priority
[    1.401902] stm32-hash 54002000.hash: Init HASH done HW ver 23 DMA mode 1
[    1.414280] remoteproc remoteproc0: releasing m4
[    1.416048] stm32-ddr-pmu: probed (ID=0x00140061 VER=0x00000010), DDR@533M
Hz
[    1.418521] /soc/interrupt-controller@5000d000: bank0
[    1.418541] /soc/interrupt-controller@5000d000: bank1
[    1.418554] /soc/interrupt-controller@5000d000: bank2
[    1.420877] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOA bank add
ed
[    1.421454] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOB bank add
ed
[    1.422006] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOC bank add
ed
[    1.422527] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOD bank add
ed
[    1.423041] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOE bank add
ed
[    1.423536] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOF bank add
ed
[    1.424034] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOG bank add
ed
[    1.424541] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOH bank add
ed
[    1.425056] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOI bank add
ed
[    1.425388] stm32mp157-pinctrl soc:pin-controller@50002000: Pinctrl STM32
initialized
[    1.426529] stm32mp157-pinctrl soc:pin-controller-z@54004000: GPIOZ bank a
dded
[    1.426559] stm32mp157-pinctrl soc:pin-controller-z@54004000: Pinctrl STM3
2 initialized
[    1.429148] stm32-usbphyc 5a006000.usbphyc: registered rev:1.0
[    1.430670] stm32-usart 40010000.serial: interrupt mode for rx (no dma)
[    1.430693] stm32-usart 40010000.serial: interrupt mode for tx (no dma)
[    1.430729] 40010000.serial: ttySTM0 at MMIO 0x40010000 (irq = 55, base_ba
ud = 4000000) is a stm32-usart
[    2.813527] printk: console [ttySTM0] enabled
[    2.819789] 44003000.serial: ttySTM1 at MMIO 0x44003000 (irq = 56, base_ba
ud = 4000000) is a stm32-usart
[    2.829496] stm32-usart 5c000000.serial: interrupt mode for rx (no dma)
[    2.834651] stm32-usart 5c000000.serial: interrupt mode for tx (no dma)
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[    2.841353] 5c000000.serial: ttySTM2 at MMIO 0x5c000000 (irq = 57, base_ba
ud = 4000000) is a stm32-usart
[    2.851029] serial serial0: tty port ttySTM2 registered
[    2.857694] m_can_platform 4400f000.can: m_can device registered (irq=52,
version=32)
[    2.865988] stm32-dwmac 5800a000.ethernet: PTP uses main clock
[    2.870434] stm32-dwmac 5800a000.ethernet: no reset control found
[    2.876480] stm32-dwmac 5800a000.ethernet: No phy clock provided...
[    2.883418] stm32-dwmac 5800a000.ethernet: User ID: 0x40, Synopsys ID: 0x4
2
[    2.889710] stm32-dwmac 5800a000.ethernet:   DWMAC4/5
[    2.894602] stm32-dwmac 5800a000.ethernet: DMA HW capability register supp
orted
[    2.901925] stm32-dwmac 5800a000.ethernet: RX Checksum Offload Engine supp
orted
[    2.909218] stm32-dwmac 5800a000.ethernet: TX Checksum insertion supported
[    2.916062] stm32-dwmac 5800a000.ethernet: Wake-Up On Lan supported
[    2.922338] stm32-dwmac 5800a000.ethernet: TSO supported
[    2.927620] stm32-dwmac 5800a000.ethernet: Enable RX Mitigation via HW Wat
chdog Timer
[    2.935467] stm32-dwmac 5800a000.ethernet: TSO feature enabled
[    2.941713] libphy: stmmac: probed
[    2.944941] yzhang..get YT8511, abt to set 125m clk out, phyaddr=6, phyid=
0000010a
[    2.953237] yt8511_config_out_125m, phy clk out, val=0x008057
[    2.958347] yzhang..8511 set 125m clk out, reg=0x8057
[    2.965537] dwc2 49000000.usb-otg: 49000000.usb-otg supply vusb_d not foun
d, using dummy regulator
[    2.973263] dwc2 49000000.usb-otg: 49000000.usb-otg supply vusb_a not foun
d, using dummy regulator
[    3.108215] dwc2 49000000.usb-otg: EPs: 9, dedicated fifos, 952 entries in
 SPRAM
[    3.115208] dwc2 49000000.usb-otg: DWC OTG Controller
[    3.119301] dwc2 49000000.usb-otg: new USB bus registered, assigned bus nu
mber 1
[    3.126628] dwc2 49000000.usb-otg: irq 59, io mem 0x49000000
[    3.132583] usb usb1: New USB device found, idVendor=1d6b, idProduct=0002,
 bcdDevice= 5.04
[    3.140513] usb usb1: New USB device strings: Mfr=3, Product=2, SerialNumb
er=1
[    3.147687] usb usb1: Product: DWC OTG Controller
[    3.152402] usb usb1: Manufacturer: Linux 5.4.31 dwc2_hsotg
[    3.157965] usb usb1: SerialNumber: 49000000.usb-otg
[    3.163847] hub 1-0:1.0: USB hub found
[    3.166693] hub 1-0:1.0: 1 port detected
[    3.172375] ehci-platform 5800d000.usbh-ehci: EHCI Host Controller
[    3.177127] ehci-platform 5800d000.usbh-ehci: new USB bus registered, assi
gned bus number 2
[    3.185965] ehci-platform 5800d000.usbh-ehci: irq 60, io mem 0x5800d000
[    3.217984] ehci-platform 5800d000.usbh-ehci: USB 2.0 started, EHCI 1.00
[    3.223657] usb usb2: New USB device found, idVendor=1d6b, idProduct=0002,
 bcdDevice= 5.04
[    3.231522] usb usb2: New USB device strings: Mfr=3, Product=2, SerialNumb
er=1
```

```
[    3.238707] usb usb2: Product: EHCI Host Controller
[    3.243546] usb usb2: Manufacturer: Linux 5.4.31 ehci_hcd
[    3.248959] usb usb2: SerialNumber: 5800d000.usbh-ehci
[    3.254981] hub 2-0:1.0: USB hub found
[    3.257859] hub 2-0:1.0: 2 ports detected
[    3.263235] ohci-platform 5800c000.usbh-ohci: Generic Platform OHCI contro
ller
[    3.269108] ohci-platform 5800c000.usbh-ohci: new USB bus registered, assi
gned bus number 3
[    3.277767] ohci-platform 5800c000.usbh-ohci: irq 51, io mem 0x5800c000
[    3.352322] usb usb3: New USB device found, idVendor=1d6b, idProduct=0001,
 bcdDevice= 5.04
[    3.359170] usb usb3: New USB device strings: Mfr=3, Product=2, SerialNumb
er=1
[    3.366332] usb usb3: Product: Generic Platform OHCI controller
[    3.372267] usb usb3: Manufacturer: Linux 5.4.31 ohci_hcd
[    3.377625] usb usb3: SerialNumber: 5800c000.usbh-ohci
[    3.384539] hub 3-0:1.0: USB hub found
[    3.386886] hub 3-0:1.0: 2 ports detected
[    3.393637] stm32_rtc 5c004000.rtc: registered as rtc0
[    3.398230] stm32_rtc 5c004000.rtc: Date/Time must be initialized
[    3.403394] stm32_rtc 5c004000.rtc: registered rev:1.2
[    3.429575] stm32f7-i2c 40012000.i2c: doesn't use DMA
[    3.435340] fusb302 0-0022: Can't get property of role, set role to defaul
t DRP
[    3.449171] fusb302 0-0022: port 0 probe success with role ROLE_MODE_DRP,
try_role ROLE_MODE_NONE
[    3.457338] stm32f7-i2c 40012000.i2c: STM32F7 I2C-0 bus adapter
[    3.483644] stm32f7-i2c 40013000.i2c: doesn't use DMA
[    3.487988] stm32f7-i2c 40013000.i2c: STM32F7 I2C-1 bus adapter
[    3.516935] stm32f7-i2c 5c002000.i2c: doesn't use DMA
[    3.522010] Goodix-TS 2-005d: 2-005d supply AVDD28 not found, using dummy
regulator
[    3.528434] Goodix-TS 2-005d: 2-005d supply VDDIO not found, using dummy r
egulator
[    3.648421] Goodix-TS 2-005d: ID 911, version: 1060
[    3.651992] stm32f7-i2c 5c002000.i2c: STM32F7 I2C-2 bus adapter
[    3.660268] mmci-pl18x 48004000.sdmmc: allocated mmc-pwrseq
[    3.664789] mmci-pl18x 48004000.sdmmc: mmc0: PL180 manf 53 rev1 at 0x48004
000 irq 42,0 (pio)
[    3.673394] Goodix-TS 2-005d: Direct firmware load for goodix_911_cfg.bin
failed with error -2
[    3.687122] input: Goodix Capacitive TouchScreen as /devices/platform/soc/
5c002000.i2c/i2c-2/2-005d/input/input0
[    3.699678] mmci-pl18x 58005000.sdmmc: mmc1: PL180 manf 53 rev2 at 0x58005
000 irq 48,0 (pio)
[    3.723658] mmc0: queuing unknown CIS tuple 0x80 (2 bytes)
[    3.729155] mmc0: queuing unknown CIS tuple 0x80 (3 bytes)
[    3.735120] mmci-pl18x 58007000.sdmmc: mmc2: PL180 manf 53 rev2 at 0x58007
000 irq 49,0 (pio)
[    3.744680] mmc0: queuing unknown CIS tuple 0x80 (3 bytes)
[    3.752468] mmc0: queuing unknown CIS tuple 0x80 (7 bytes)
[    3.762563] mmc0: queuing unknown CIS tuple 0x81 (9 bytes)
[    3.774364] stm32-ipcc 4c001000.mailbox: ipcc rev:1.0 enabled, 6 chans, pr
```

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oc 0
[    3.784562] stm32-rproc mlahb:m4@10000000: wdg irq registered
[    3.791086] remoteproc remoteproc0: m4 is available
[    3.796944] stm32mp157-pinctrl soc:pin-controller@50002000: missing pins p
roperty in node pins .
[    3.804460] stm32-adc-core: probe of 48003000.adc failed with error -22
[    3.889196] mmc2: new high speed MMC card at address 0001
[    3.905596] mmcblk2: mmc2:0001 S40004 3.64 GiB
[    3.909480] mmcblk2boot0: mmc2:0001 S40004 partition 1 4.00 MiB
[    3.915275] mmcblk2boot1: mmc2:0001 S40004 partition 2 4.00 MiB
[    3.925220] mmc0: new high speed SDIO card at address 0001
[    3.930271] mmcblk2rpmb: mmc2:0001 S40004 partition 3 4.00 MiB, chardev (2
42:0)
[    3.958397] asoc-audio-graph-card sound: wm8960-hifi-Playback <-> 4400a004
.audio-controller mapping ok
[    3.966697]  mmcblk2: p1 p2 p3
[    3.974569] asoc-audio-graph-card sound: wm8960-hifi-Capture <-> 4400a024.
audio-controller mapping ok
[    4.105176] input: joystick as /devices/platform/joystick/input/input1
[    4.110908] stm32_rtc 5c004000.rtc: setting system clock to 2000-01-01T10:
17:34 UTC (946721854)
[    4.120309] cfg80211: Loading compiled-in X.509 certificates for regulator
y database
[    4.129761] cfg80211: Loaded X.509 cert 'sforshee: 00b28ddf47aef9cea7'
[    4.135071] platform regulatory.0: Direct firmware load for regulatory.db
failed with error -2
[    4.144679] ALSA device list:
[    4.146378]   #0: WM8960-100ASK
[    4.149583] cfg80211: failed to load regulatory.db
[    4.156131] Waiting for root device /dev/mmcblk1p3...
[   34.407980] vmmcsd_fixed: disabling



pwd
 ^H^H^H^C^[^[^[^[
reboot
```

```
NOTICE:  CPU: STM32MP157AAC Rev.Z
NOTICE:  Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
INFO:    Reset reason (0x15):
INFO:      Power-on Reset (rst_por)
INFO:    Using EMMC
INFO:      Instance 2
INFO:    Boot used partition fsbl1
NOTICE:  BL2: v2.2-r1.0(debug):85b7ab8
NOTICE:  BL2: Built : 09:15:17, Aug 16 2026
INFO:    Using crypto library 'stm32_crypto_lib'
INFO:    BL2: Doing platform setup
INFO:    RAM: DDR3-1066/888 bin G 2x2Gb 533MHz v1.44
INFO:    Memory size = 0x20000000 (512 MB)
INFO:    BL2 runs SP_MIN setup
INFO:    BL2: Loading image id 4
INFO:    Loading image id=4 at address 0x2ffed000
INFO:    Image id=4 loaded: 0x2ffed000 - 0x2ffff000
INFO:    BL2: Loading image id 5
INFO:    Loading image id=5 at address 0xc0100000
INFO:    STM32 Image size : 869897
INFO:    Image id=5 loaded: 0xc0100000 - 0xc01d4609
WARNING: Skip signature check (header option)
NOTICE:  ROTPK is not deployed on platform. Skipping ROTPK verification.
NOTICE:  BL2: Booting BL32
INFO:    Entry point address = 0x2ffed000
INFO:    SPSR = 0x1d3
NOTICE:  SP_MIN: v2.2-r1.0(debug):85b7ab8
NOTICE:  SP_MIN: Built : 09:15:17, Aug 16 2026
INFO:    ARM GICv2 driver initialized
INFO:    stm32mp IWDG1 (12): Secure
INFO:    ETZPC: CRYP1 (9) could be non secure
INFO:    SP_MIN: Initializing runtime services
INFO:    SP_MIN: Preparing exit to normal world


U-Boot 2020.01-stm32mp-r1 (Aug 16 2026 - 09:15:35 +0800)

CPU: STM32MP157AAC Rev.Z
Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
Board: stm32mp1 in trusted mode (st,stm32mp157c-100ask-512d-v1)
DRAM:  512 MiB
Clocks:
- MPU : 650 MHz
- MCU : 208.878 MHz
- AXI : 266.500 MHz
- PER : 24 MHz
- DDR : 533 MHz
WDT:   Started with servicing (32s timeout)
```

```
NAND:  0 MiB
MMC:   STM32 SD/MMC: 0, STM32 SD/MMC: 1
Loading Environment from MMC... OK
In:    serial
Out:   serial
Err:   serial
Net:   eth0: ethernet@5800a000
Press SPACE to abort autoboot in 1 seconds
Boot over mmc1!
switch to partitions #0, OK
mmc1(part 0) is current device
Scanning mmc 1:2...
Found /mmc1_extlinux/stm32mp157c-100ask-512d-v1_extlinux.conf
Retrieving file: /mmc1_extlinux/stm32mp157c-100ask-512d-v1_extlinux.conf
210 bytes read in 37 ms (4.9 KiB/s)
1:      100ask-lcd
Retrieving file: /uImage
8171336 bytes read in 315 ms (24.7 MiB/s)
append: root=/dev/mmcblk2p3 rootwait rw console=ttySTM0,115200
Retrieving file: /stm32mp157c-100ask-512d-lcd-v1.dtb
74438 bytes read in 40 ms (1.8 MiB/s)
## Booting kernel from Legacy Image at c2000000 ...
   Image Name:   Linux-5.4.31-g893d15a84
   Created:      2026-10-21  11:38:18 UTC
   Image Type:   ARM Linux Kernel Image (uncompressed)
   Data Size:    8171272 Bytes = 7.8 MiB
   Load Address: 0c200040
   Entry Point:  0c200040
   Verifying Checksum ... OK
## Flattened Device Tree blob at c4000000
   Booting using the fdt blob at 0xc4000000
   Loading Kernel Image
   Loading Device Tree to cffea000, end cffff2c5 ... OK

Starting kernel ...

prefetch abort
pc : [<0c20003c>]          lr : [<d9c47573>]
reloc pc : [<f26ba03c>]    lr : [<c0101573>]
sp : d7aeb278  ip : d7c45102     fp : c2000000
r10: d9c47695  r9 : d7affec0     r8 : d7c5a1c8
r7 : 00000000  r6 : 0c200040     r5 : d9cf4030  r4 : 00000000
r3 : 000122c6  r2 : cffea000     r1 : 00000000  r0 : 00000000
Flags: nZCv  IRQs off  FIQs off  Mode SVC_32
Code: data abort
pc : [<d9c47920>]          lr : [<d9c4790d>]
reloc pc : [<c0101920>]    lr : [<c010190d>]
sp : d7aeb180  ip : 0000003c     fp : c2000000
r10: d9cd478a  r9 : d7affec0     r8 : d9cd4782
r7 : 00000000  r6 : 00000008     r5 : 0c20003c  r4 : fffffffc
r3 : 00000001  r2 : 00000028     r1 : 00000000  r0 : 00000006
Flags: NzCv  IRQs on  FIQs on  Mode SVC_32 (T)
Code: 2f00 d1e9 2c00 dce9 (f855) 2024
Resetting CPU ...
```

```
resetting ...
INFO:    PSCI Power Domain Map:
INFO:      Domain Node : Level 1, parent_node -1, State ON (0x0)
INFO:      Domain Node : Level 0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0x0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0xffffffff, parent_node 0, State OFF (0x2)
NOTICE:  CPU: STM32MP157AAC Rev.Z
NOTICE:  Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
INFO:    Reset reason (0x54):
INFO:      System reset generated by MPU (MPSYSRST)
INFO:    Using EMMC
INFO:      Instance 2
INFO:    Boot used partition fsbl1
NOTICE:  BL2: v2.2-r1.0(debug):85b7ab8
NOTICE:  BL2: Built : 09:15:17, Aug 16 2026
INFO:    Using crypto library 'stm32_crypto_lib'
INFO:    BL2: Doing platform setup
INFO:    RAM: DDR3-1066/888 bin G 2x2Gb 533MHz v1.44
INFO:    Memory size = 0x20000000 (512 MB)
INFO:    BL2 runs SP_MIN setup
INFO:    BL2: Loading image id 4
INFO:    Loading image id=4 at address 0x2ffed000
INFO:    Image id=4 loaded: 0x2ffed000 - 0x2ffff000
INFO:    BL2: Loading image id 5
INFO:    Loading image id=5 at address 0xc0100000
INFO:    STM32 Image size : 869897
INFO:    Image id=5 loaded: 0xc0100000 - 0xc01d4609
WARNING: Skip signature check (header option)
NOTICE:  ROTPK is not deployed on platform. Skipping ROTPK verification.
NOTICE:  BL2: Booting BL32
INFO:    Entry point address = 0x2ffed000
INFO:    SPSR = 0x1d3
NOTICE:  SP_MIN: v2.2-r1.0(debug):85b7ab8
NOTICE:  SP_MIN: Built : 09:15:17, Aug 16 2026
INFO:    ARM GICv2 driver initialized
INFO:    stm32mp IWDG1 (12): Secure
INFO:    ETZPC: CRYP1 (9) could be non secure
INFO:    SP_MIN: Initializing runtime services
INFO:    SP_MIN: Preparing exit to normal world


U-Boot 2020.01-stm32mp-r1 (Aug 16 2026 - 09:15:35 +0800)

CPU: STM32MP157AAC Rev.Z
Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
Board: stm32mp1 in trusted mode (st,stm32mp157c-100ask-512d-v1)
DRAM:  512 MiB
Clocks:
- MPU : 650 MHz
- MCU : 208.878 MHz
- AXI : 266.500 MHz
- PER : 24 MHz
- DDR : 533 MHz
```

```
WDT:   Started with servicing (32s timeout)
NAND:  0 MiB
MMC:   STM32 SD/MMC: 0, STM32 SD/MMC: 1
Loading Environment from MMC... OK
In:    serial
Out:   serial
Err:   serial
Net:   eth0: ethernet@5800a000
Press SPACE to abort autoboot in 1 seconds
STM32MP> tftp 0xc4200000 uImage
yt8511_config_out_125m, phy clk out, vall=8057
ethernet@5800a000 Waiting for PHY auto negotiation to complete... done
Using ethernet@5800a000 device
TFTP from server 192.168.145.128; our IP address is 192.168.145.129
Filename 'uImage'.
Load address: 0xc4200000
Loading: #################################################################
         #################################################################
         #################################################################
         ##############################################T ##################
         #################################################################
         #################################################################
         #################################################################
         #################################################################
         ######################################
         296.9 KiB/s
done
Bytes transferred = 8187672 (7cef18 hex)
STM32MP> tftp 0xc4000000 stm32mp157c-100ask-512d-v1.dtb
Using ethernet@5800a000 device
TFTP from server 192.168.145.128; our IP address is 192.168.145.129
Filename 'stm32mp157c-100ask-512d-v1.dtb'.
Load address: 0xc4000000
Loading: ########
         635.7 KiB/s
done
Bytes transferred = 107530 (1a40a hex)
STM32MP> setenv bootargs 'root=/dev/mmcblk2p3 rootwait rw console=ttySTM0,115
200'
STM32MP> bootm 0xc4200000 - 0xc4000000
## Booting kernel from Legacy Image at c4200000 ...
   Image Name:   Linux-5.4.31
   Created:      2022-10-27  11:09:36 UTC
   Image Type:   ARM Linux Kernel Image (uncompressed)
   Data Size:    8187608 Bytes = 7.8 MiB
   Load Address: c2000040
   Entry Point:  c2000040
   Verifying Checksum ... OK
## Flattened Device Tree blob at c4000000
   Booting using the fdt blob at 0xc4000000
   Loading Kernel Image
   Loading Device Tree to cffe2000, end cffff409 ... OK

Starting kernel ...
```

```
[    0.000000] Booting Linux on physical CPU 0x0
[    0.000000] Linux version 5.4.31 (book@virtual-machine) (gcc version 8.4.0
 (Buildroot 2020.02-g8ad4a8c-dirty)) #1 SMP PREEMPT Thu Oct 27 11:04:26 UTC 2
022
[    0.000000] CPU: ARMv7 Processor [410fc075] revision 5 (ARMv7), cr=10c5387
d
[    0.000000] CPU: div instructions available: patching division code
[    0.000000] CPU: PIPT / VIPT nonaliasing data cache, VIPT aliasing instruc
tion cache
[    0.000000] OF: fdt: Machine model: 100ASK YA157C v2 www.100ask.com
[    0.000000] Memory policy: Data cache writealloc
[    0.000000] Reserved memory: created DMA memory pool at 0x10000000, size 0
 MiB
[    0.000000] OF: reserved mem: initialized node mcuram2@10000000, compatibl
e id shared-dma-pool
[    0.000000] Reserved memory: created DMA memory pool at 0x10040000, size 0
 MiB
[    0.000000] OF: reserved mem: initialized node vdev0vring0@10040000, compa
tible id shared-dma-pool
[    0.000000] Reserved memory: created DMA memory pool at 0x10041000, size 0
 MiB
[    0.000000] OF: reserved mem: initialized node vdev0vring1@10041000, compa
tible id shared-dma-pool
[    0.000000] Reserved memory: created DMA memory pool at 0x10042000, size 0
 MiB
[    0.000000] OF: reserved mem: initialized node vdev0buffer@10042000, compa
tible id shared-dma-pool
[    0.000000] Reserved memory: created DMA memory pool at 0x30000000, size 0
 MiB
[    0.000000] OF: reserved mem: initialized node mcuram@30000000, compatible
 id shared-dma-pool
[    0.000000] Reserved memory: created DMA memory pool at 0x38000000, size 0
 MiB
[    0.000000] OF: reserved mem: initialized node retram@38000000, compatible
 id shared-dma-pool
[    0.000000] cma: Reserved 128 MiB at 0xd2000000
[    0.000000] psci: probing for conduit method from DT.
[    0.000000] psci: PSCIv1.1 detected in firmware.
[    0.000000] psci: Using standard PSCI v0.2 function IDs
[    0.000000] psci: MIGRATE_INFO_TYPE not supported.
[    0.000000] psci: SMC Calling Convention v1.0
[    0.000000] percpu: Embedded 20 pages/cpu s50124 r8192 d23604 u81920
[    0.000000] Built 1 zonelists, mobility grouping on.  Total pages: 113536
[    0.000000] Kernel command line: root=/dev/mmcblk2p3 rootwait rw console=t
tySTM0,115200
[    0.000000] Dentry cache hash table entries: 65536 (order: 6, 262144 bytes
, linear)
[    0.000000] Inode-cache hash table entries: 32768 (order: 5, 131072 bytes,
 linear)
[    0.000000] mem auto-init: stack:off, heap alloc:off, heap free:off
[    0.000000] Memory: 303816K/458752K available (12288K kernel code, 962K rw
data, 3516K rodata, 1024K init, 226K bss, 23864K reserved, 131072K cma-reserv
ed, 0K highmem)
```

```
[    0.000000] SLUB: HWalign=64, Order=0-3, MinObjects=0, CPUs=2, Nodes=1
[    0.000000] rcu: Preemptible hierarchical RCU implementation.
[    0.000000] rcu:     RCU event tracing is enabled.
[    0.000000]  Tasks RCU enabled.
[    0.000000] rcu: RCU calculated value of scheduler-enlistment delay is 10
jiffies.
[    0.000000] NR_IRQS: 16, nr_irqs: 16, preallocated irqs: 16
[    0.000000] random: get_random_bytes called from start_kernel+0x320/0x4f4
with crng_init=0
[    0.000000] arch_timer: cp15 timer(s) running at 24.00MHz (virt).
[    0.000000] clocksource: arch_sys_counter: mask: 0xffffffffffffff max_cycl
es: 0x588fe9dc0, max_idle_ns: 440795202592 ns
[    0.000009] sched_clock: 56 bits at 24MHz, resolution 41ns, wraps every 43
98046511097ns
[    0.000027] Switching to timer-based delay loop, resolution 41ns
[    0.001995] Console: colour dummy device 80x30
[    0.002048] Calibrating delay loop (skipped), value calculated using timer
 frequency.. 48.00 BogoMIPS (lpj=240000)
[    0.002069] pid_max: default: 32768 minimum: 301
[    0.002331] Mount-cache hash table entries: 1024 (order: 0, 4096 bytes, li
near)
[    0.002355] Mountpoint-cache hash table entries: 1024 (order: 0, 4096 byte
s, linear)
[    0.003650] CPU: Testing write buffer coherency: ok
[    0.004187] /cpus/cpu@0 missing clock-frequency property
[    0.004224] /cpus/cpu@1 missing clock-frequency property
[    0.004242] CPU0: thread -1, cpu 0, socket 0, mpidr 80000000
[    0.059760] Setting up static identity map for 0xc0100000 - 0xc0100060
[    0.079736] rcu: Hierarchical SRCU implementation.
[    0.105118] smp: Bringing up secondary CPUs ...
[    0.180375] CPU1: thread -1, cpu 1, socket 0, mpidr 80000001
[    0.180596] smp: Brought up 1 node, 2 CPUs
[    0.180620] SMP: Total of 2 processors activated (96.00 BogoMIPS).
[    0.180631] CPU: All CPU(s) started in SVC mode.
[    0.181477] devtmpfs: initialized
[    0.214396] VFP support v0.3: implementor 41 architecture 2 part 30 varian
t 7 rev 5
[    0.215020] clocksource: jiffies: mask: 0xffffffff max_cycles: 0xffffffff,
 max_idle_ns: 19112604462750000 ns
[    0.215054] futex hash table entries: 512 (order: 3, 32768 bytes, linear)
[    0.221958] pinctrl core: initialized pinctrl subsystem
[    0.224370] NET: Registered protocol family 16
[    0.233119] DMA: preallocated 256 KiB pool for atomic coherent allocations
[    0.235715] hw-breakpoint: found 5 (+1 reserved) breakpoint and 4 watchpoi
nt registers.
[    0.235737] hw-breakpoint: maximum watchpoint size is 8 bytes.
[    0.236164] Serial: AMBA PL011 UART driver
[    0.239385] stm32-pm-domain pm_domain: domain core-ret-power-domain regist
ered
[    0.239446] stm32-pm-domain pm_domain: subdomain core-power-domain registe
red
[    0.239463] stm32-pm-domain pm_domain: domains probed
[    0.291200] gpio-regulator regulators: No 'states' property found
[    0.291243] gpio-regulator: probe of regulators failed with error -22
```

```
[    0.292686] SCSI subsystem initialized
[    0.293533] usbcore: registered new interface driver usbfs
[    0.293617] usbcore: registered new interface driver hub
[    0.293779] usbcore: registered new device driver usb
[    0.294168] pps_core: LinuxPPS API ver. 1 registered
[    0.294182] pps_core: Software ver. 5.3.6 - Copyright 2005-2007 Rodolfo Gi
ometti <giometti@linux.it>
[    0.294215] PTP clock support registered
[    0.294570] EDAC MC: Ver: 3.0.0
[    0.295894] Advanced Linux Sound Architecture Driver Initialized.
[    0.296753] Bluetooth: Core ver 2.22
[    0.296860] NET: Registered protocol family 31
[    0.296873] Bluetooth: HCI device and connection manager initialized
[    0.296897] Bluetooth: HCI socket layer initialized
[    0.296914] Bluetooth: L2CAP socket layer initialized
[    0.296950] Bluetooth: SCO socket layer initialized
[    0.297938] clocksource: Switched to clocksource arch_sys_counter
[    1.057571] thermal_sys: Registered thermal governor 'step_wise'
[    1.058287] NET: Registered protocol family 2
[    1.059394] tcp_listen_portaddr_hash hash table entries: 512 (order: 0, 61
44 bytes, linear)
[    1.059450] TCP established hash table entries: 4096 (order: 2, 16384 byte
s, linear)
[    1.059529] TCP bind hash table entries: 4096 (order: 3, 32768 bytes, line
ar)
[    1.059626] TCP: Hash tables configured (established 4096 bind 4096)
[    1.059772] UDP hash table entries: 256 (order: 1, 8192 bytes, linear)
[    1.059818] UDP-Lite hash table entries: 256 (order: 1, 8192 bytes, linear
)
[    1.060159] NET: Registered protocol family 1
[    1.061177] RPC: Registered named UNIX socket transport module.
[    1.061197] RPC: Registered udp transport module.
[    1.061208] RPC: Registered tcp transport module.
[    1.061218] RPC: Registered tcp NFSv4.1 backchannel transport module.
[    1.063048] hw perfevents: enabled with armv7_cortex_a7 PMU driver, 5 coun
ters available
[    1.065147] Initialise system trusted keyrings
[    1.065569] workingset: timestamp_bits=14 max_order=17 bucket_order=3
[    1.078259] squashfs: version 4.0 (2009/01/31) Phillip Lougher
[    1.079611] NFS: Registering the id_resolver key type
[    1.079664] Key type id_resolver registered
[    1.079676] Key type id_legacy registered
[    1.079915] ntfs: driver 2.1.32 [Flags: R/O].
[    1.080425] jffs2: version 2.2. (NAND) © 2001-2006 Red Hat, Inc.
[    1.081217] fuse: init (API version 7.31)
[    1.139215] NET: Registered protocol family 38
[    1.139245] Key type asymmetric registered
[    1.139258] Asymmetric key parser 'x509' registered
[    1.139351] Block layer SCSI generic (bsg) driver version 0.4 loaded (majo
r 246)
[    1.139367] io scheduler mq-deadline registered
[    1.139379] io scheduler kyber registered
[    1.149824] STM32 USART driver initialized
[    1.151404] ar8035 phy init
```

```
[    1.151875] the phy gpio failed
[    1.172718] brd: module loaded
[    1.187074] loop: module loaded
[    1.193423] libphy: Fixed MDIO Bus: probed
[    1.195307] CAN device driver interface
[    1.197789] PPP generic driver version 2.4.2
[    1.198235] PPP BSD Compression module registered
[    1.198252] PPP Deflate Compression module registered
[    1.198279] PPP MPPE Compression module registered
[    1.198292] NET: Registered protocol family 24
[    1.198339] SLIP: version 0.8.4-NET3.019-NEWTTY (dynamic channels, max=256
) (6 bit encapsulation enabled).
[    1.198350] CSLIP: code copyright 1989 Regents of the University of Califo
rnia.
[    1.198359] SLIP linefill/keepalive option.
[    1.198374] pegasus: v0.9.3 (2013/04/25), Pegasus/Pegasus II USB Ethernet
driver
[    1.198479] usbcore: registered new interface driver pegasus
[    1.198580] usbcore: registered new interface driver r8152
[    1.198656] usbcore: registered new interface driver asix
[    1.198715] usbcore: registered new interface driver ax88179_178a
[    1.198775] usbcore: registered new interface driver cdc_ether
[    1.198853] usbcore: registered new interface driver smsc75xx
[    1.198931] usbcore: registered new interface driver smsc95xx
[    1.199002] usbcore: registered new interface driver net1080
[    1.199062] usbcore: registered new interface driver cdc_subset
[    1.199119] usbcore: registered new interface driver zaurus
[    1.199210] usbcore: registered new interface driver cdc_ncm
[    1.200875] ehci_hcd: USB 2.0 'Enhanced' Host Controller (EHCI) Driver
[    1.200935] ehci-platform: EHCI generic platform driver
[    1.201442] ohci_hcd: USB 1.1 'Open' Host Controller (OHCI) Driver
[    1.201482] ohci-platform: OHCI generic platform driver
[    1.202498] usbcore: registered new interface driver cdc_acm
[    1.202513] cdc_acm: USB Abstract Control Model driver for USB modems and
ISDN adapters
[    1.202598] usbcore: registered new interface driver usblp
[    1.202660] usbcore: registered new interface driver cdc_wdm
[    1.202718] usbcore: registered new interface driver uas
[    1.202821] usbcore: registered new interface driver usb-storage
[    1.202974] usbcore: registered new interface driver usbserial_generic
[    1.203026] usbserial: USB Serial support registered for generic
[    1.203083] usbcore: registered new interface driver option
[    1.203126] usbserial: USB Serial support registered for GSM modem (1-port
)
[    1.208240] i2c /dev entries driver
[    1.213169] stm32-cpufreq stm32-cpufreq: Failed to get chip info: -517
[    1.217732] sdhci: Secure Digital Host Controller Interface driver
[    1.217748] sdhci: Copyright(c) Pierre Ossman
[    1.217759] Synopsys Designware Multimedia Card Interface Driver
[    1.218466] sdhci-pltfm: SDHCI platform and OF driver helper
[    1.219751] ledtrig-cpu: registered to indicate activity on CPUs
[    1.222083] usbcore: registered new interface driver usbhid
[    1.222103] usbhid: USB HID core driver
[    1.222113] dtbocfg_module_init
```

```
[    1.222166] dtbocfg_module_init: OK
[    1.222537] arm-smc-mbox mailbox-0: ARM SMC mailbox enabled.
[    1.222672] arm-smc-mbox mailbox-1: ARM SMC mailbox enabled.
[    1.224398] remoteproc remoteproc0: releasing m4
[    1.227521] Unable to get STM32 DDR PMU clock
[    1.232953] drop_monitor: Initializing network drop monitor service
[    1.233307] NET: Registered protocol family 17
[    1.233342] can: controller area network core (rev 20170425 abi 9)
[    1.233471] NET: Registered protocol family 29
[    1.233488] can: raw protocol (rev 20170425)
[    1.233500] can: broadcast manager protocol (rev 20170425 t)
[    1.233520] can: netlink gateway (rev 20190810) max_hops=1
[    1.234026] Bluetooth: RFCOMM TTY layer initialized
[    1.234061] Bluetooth: RFCOMM socket layer initialized
[    1.234114] Bluetooth: RFCOMM ver 1.11
[    1.234143] Bluetooth: BNEP (Ethernet Emulation) ver 1.3
[    1.234154] Bluetooth: BNEP filters: protocol multicast
[    1.234175] Bluetooth: BNEP socket layer initialized
[    1.234187] Bluetooth: HIDP (Human Interface Emulation) ver 1.2
[    1.234204] Bluetooth: HIDP socket layer initialized
[    1.234776] Key type dns_resolver registered
[    1.234973] ThumbEE CPU extension supported.
[    1.234999] Registering SWP/SWPB emulation handler
[    1.235462] registered taskstats version 1
[    1.235478] Loading compiled-in X.509 certificates
[    1.246753] Loaded X.509 cert 'Build time autogenerated kernel key: a551a3
bcaf9a62271059adf2e1003b2f3df06de0'
[    1.271359] arm-scmi firmware:scmi-0: SCMI Protocol v2.0 'ST:' Firmware ve
rsion 0x0
[    1.275689] remoteproc remoteproc0: releasing m4
[    1.277139] Unable to get STM32 DDR PMU clock
[    1.298207] debugfs: Directory 'cpu0' with parent 'opp' already present!
[    1.307894] remoteproc remoteproc0: releasing m4
[    1.309462] Unable to get STM32 DDR PMU clock
[    1.348371] stm32-mdma 58000000.dma: STM32 MDMA driver registered
[    1.352902] stm32-dma 48000000.dma: STM32 DMA driver registered
[    1.355022] stm32-dma 48001000.dma: STM32 DMA driver registered
[    1.355697] reg11: supplied by regulator-dummy
[    1.356160] reg18: supplied by regulator-dummy
[    1.356573] usb33: supplied by regulator-dummy
[    1.378375] stm_thermal 50028000.thermal: stm_thermal_probe: Driver initia
lized successfully
[    1.381226] debugfs: Directory 'cpu0' with parent 'opp' already present!
[    1.381528] random: fast init done
[    1.385348] stm32-crc32 58009000.crc: Initialized
[    1.385944] random: crng init done
[    1.386975] stm32-hash 54002000.hash: will run requests pump with realtime
 priority
[    1.391964] stm32-hash 54002000.hash: Init HASH done HW ver 23 DMA mode 1
[    1.404344] remoteproc remoteproc0: releasing m4
[    1.406132] stm32-ddr-pmu: probed (ID=0x00140061 VER=0x00000010), DDR@533M
Hz
[    1.408639] /soc/interrupt-controller@5000d000: bank0
[    1.408662] /soc/interrupt-controller@5000d000: bank1
```

```
[    1.408674] /soc/interrupt-controller@5000d000: bank2
[    1.411011] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOA bank add
ed
[    1.411615] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOB bank add
ed
[    1.412141] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOC bank add
ed
[    1.412679] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOD bank add
ed
[    1.413197] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOE bank add
ed
[    1.413705] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOF bank add
ed
[    1.414228] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOG bank add
ed
[    1.414737] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOH bank add
ed
[    1.415253] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOI bank add
ed
[    1.415586] stm32mp157-pinctrl soc:pin-controller@50002000: Pinctrl STM32
initialized
[    1.416672] stm32mp157-pinctrl soc:pin-controller-z@54004000: GPIOZ bank a
dded
[    1.416704] stm32mp157-pinctrl soc:pin-controller-z@54004000: Pinctrl STM3
2 initialized
[    1.419249] stm32-usbphyc 5a006000.usbphyc: registered rev:1.0
[    1.420775] stm32-usart 40010000.serial: interrupt mode for rx (no dma)
[    1.420797] stm32-usart 40010000.serial: interrupt mode for tx (no dma)
[    1.420833] 40010000.serial: ttySTM0 at MMIO 0x40010000 (irq = 55, base_ba
ud = 4000000) is a stm32-usart
[    2.803955] printk: console [ttySTM0] enabled
[    2.810209] 44003000.serial: ttySTM1 at MMIO 0x44003000 (irq = 56, base_ba
ud = 4000000) is a stm32-usart
[    2.819900] stm32-usart 5c000000.serial: interrupt mode for rx (no dma)
[    2.825054] stm32-usart 5c000000.serial: interrupt mode for tx (no dma)
[    2.831761] 5c000000.serial: ttySTM2 at MMIO 0x5c000000 (irq = 57, base_ba
ud = 4000000) is a stm32-usart
[    2.841414] serial serial0: tty port ttySTM2 registered
[    2.848152] m_can_platform 4400f000.can: m_can device registered (irq=52,
version=32)
[    2.856378] stm32-dwmac 5800a000.ethernet: PTP uses main clock
[    2.860825] stm32-dwmac 5800a000.ethernet: no reset control found
[    2.866869] stm32-dwmac 5800a000.ethernet: No phy clock provided...
[    2.873808] stm32-dwmac 5800a000.ethernet: User ID: 0x40, Synopsys ID: 0x4
2
[    2.880097] stm32-dwmac 5800a000.ethernet:   DWMAC4/5
[    2.884992] stm32-dwmac 5800a000.ethernet: DMA HW capability register supp
orted
[    2.892317] stm32-dwmac 5800a000.ethernet: RX Checksum Offload Engine supp
orted
[    2.899609] stm32-dwmac 5800a000.ethernet: TX Checksum insertion supported
[    2.906453] stm32-dwmac 5800a000.ethernet: Wake-Up On Lan supported
[    2.912730] stm32-dwmac 5800a000.ethernet: TSO supported
[    2.918034] stm32-dwmac 5800a000.ethernet: Enable RX Mitigation via HW Wat
```

```
chdog Timer
[    2.925834] stm32-dwmac 5800a000.ethernet: TSO feature enabled
[    2.932117] libphy: stmmac: probed
[    2.935333] yzhang..get YT8511, abt to set 125m clk out, phyaddr=6, phyid=
0000010a
[    2.943635] yt8511_config_out_125m, phy clk out, val=0x008057
[    2.948737] yzhang..8511 set 125m clk out, reg=0x8057
[    2.955906] dwc2 49000000.usb-otg: 49000000.usb-otg supply vusb_d not foun
d, using dummy regulator
[    2.963672] dwc2 49000000.usb-otg: 49000000.usb-otg supply vusb_a not foun
d, using dummy regulator
[    3.098227] dwc2 49000000.usb-otg: EPs: 9, dedicated fifos, 952 entries in
 SPRAM
[    3.105182] dwc2 49000000.usb-otg: DWC OTG Controller
[    3.109318] dwc2 49000000.usb-otg: new USB bus registered, assigned bus nu
mber 1
[    3.116637] dwc2 49000000.usb-otg: irq 59, io mem 0x49000000
[    3.122592] usb usb1: New USB device found, idVendor=1d6b, idProduct=0002,
 bcdDevice= 5.04
[    3.130536] usb usb1: New USB device strings: Mfr=3, Product=2, SerialNumb
er=1
[    3.137702] usb usb1: Product: DWC OTG Controller
[    3.142418] usb usb1: Manufacturer: Linux 5.4.31 dwc2_hsotg
[    3.147980] usb usb1: SerialNumber: 49000000.usb-otg
[    3.153877] hub 1-0:1.0: USB hub found
[    3.156720] hub 1-0:1.0: 1 port detected
[    3.162396] ehci-platform 5800d000.usbh-ehci: EHCI Host Controller
[    3.167149] ehci-platform 5800d000.usbh-ehci: new USB bus registered, assi
gned bus number 2
[    3.175989] ehci-platform 5800d000.usbh-ehci: irq 60, io mem 0x5800d000
[    3.207986] ehci-platform 5800d000.usbh-ehci: USB 2.0 started, EHCI 1.00
[    3.213667] usb usb2: New USB device found, idVendor=1d6b, idProduct=0002,
 bcdDevice= 5.04
[    3.221532] usb usb2: New USB device strings: Mfr=3, Product=2, SerialNumb
er=1
[    3.228714] usb usb2: Product: EHCI Host Controller
[    3.233555] usb usb2: Manufacturer: Linux 5.4.31 ehci_hcd
[    3.238968] usb usb2: SerialNumber: 5800d000.usbh-ehci
[    3.244965] hub 2-0:1.0: USB hub found
[    3.247867] hub 2-0:1.0: 2 ports detected
[    3.253258] ohci-platform 5800c000.usbh-ohci: Generic Platform OHCI contro
ller
[    3.259127] ohci-platform 5800c000.usbh-ohci: new USB bus registered, assi
gned bus number 3
[    3.267793] ohci-platform 5800c000.usbh-ohci: irq 51, io mem 0x5800c000
[    3.342345] usb usb3: New USB device found, idVendor=1d6b, idProduct=0001,
 bcdDevice= 5.04
[    3.349194] usb usb3: New USB device strings: Mfr=3, Product=2, SerialNumb
er=1
[    3.356356] usb usb3: Product: Generic Platform OHCI controller
[    3.362291] usb usb3: Manufacturer: Linux 5.4.31 ohci_hcd
[    3.367649] usb usb3: SerialNumber: 5800c000.usbh-ohci
[    3.374552] hub 3-0:1.0: USB hub found
[    3.376891] hub 3-0:1.0: 2 ports detected
```

```
[    3.383618] stm32_rtc 5c004000.rtc: registered as rtc0
[    3.388181] stm32_rtc 5c004000.rtc: Date/Time must be initialized
[    3.393378] stm32_rtc 5c004000.rtc: registered rev:1.2
[    3.419577] stm32f7-i2c 40012000.i2c: doesn't use DMA
[    3.425319] fusb302 0-0022: Can't get property of role, set role to defaul
t DRP
[    3.439048] fusb302 0-0022: port 0 probe success with role ROLE_MODE_DRP,
try_role ROLE_MODE_NONE
[    3.447212] stm32f7-i2c 40012000.i2c: STM32F7 I2C-0 bus adapter
[    3.473532] stm32f7-i2c 40013000.i2c: doesn't use DMA
[    3.477822] stm32f7-i2c 40013000.i2c: STM32F7 I2C-1 bus adapter
[    3.506871] stm32f7-i2c 5c002000.i2c: doesn't use DMA
[    3.511933] Goodix-TS 2-005d: 2-005d supply AVDD28 not found, using dummy
regulator
[    3.518356] Goodix-TS 2-005d: 2-005d supply VDDIO not found, using dummy r
egulator
[    3.638420] Goodix-TS 2-005d: ID 911, version: 1060
[    3.641992] stm32f7-i2c 5c002000.i2c: STM32F7 I2C-2 bus adapter
[    3.650300] mmci-pl18x 48004000.sdmmc: allocated mmc-pwrseq
[    3.654844] mmci-pl18x 48004000.sdmmc: mmc0: PL180 manf 53 rev1 at 0x48004
000 irq 42,0 (pio)
[    3.663422] Goodix-TS 2-005d: Direct firmware load for goodix_911_cfg.bin
failed with error -2
[    3.677148] input: Goodix Capacitive TouchScreen as /devices/platform/soc/
5c002000.i2c/i2c-2/2-005d/input/input0
[    3.689646] mmci-pl18x 58005000.sdmmc: mmc1: PL180 manf 53 rev2 at 0x58005
000 irq 48,0 (pio)
[    3.713653] mmc0: queuing unknown CIS tuple 0x80 (2 bytes)
[    3.719157] mmc0: queuing unknown CIS tuple 0x80 (3 bytes)
[    3.725185] mmci-pl18x 58007000.sdmmc: mmc2: PL180 manf 53 rev2 at 0x58007
000 irq 49,0 (pio)
[    3.734758] mmc0: queuing unknown CIS tuple 0x80 (3 bytes)
[    3.742549] mmc0: queuing unknown CIS tuple 0x80 (7 bytes)
[    3.752613] mmc0: queuing unknown CIS tuple 0x81 (9 bytes)
[    3.764462] stm32-ipcc 4c001000.mailbox: ipcc rev:1.0 enabled, 6 chans, pr
oc 0
[    3.774613] stm32-rproc mlahb:m4@10000000: wdg irq registered
[    3.781105] remoteproc remoteproc0: m4 is available
[    3.787006] stm32mp157-pinctrl soc:pin-controller@50002000: missing pins p
roperty in node pins .
[    3.794528] stm32-adc-core: probe of 48003000.adc failed with error -22
[    3.879290] mmc2: new high speed MMC card at address 0001
[    3.895720] mmcblk2: mmc2:0001 S40004 3.64 GiB
[    3.899594] mmcblk2boot0: mmc2:0001 S40004 partition 1 4.00 MiB
[    3.905385] mmcblk2boot1: mmc2:0001 S40004 partition 2 4.00 MiB
[    3.915350] mmc0: new high speed SDIO card at address 0001
[    3.920244] mmcblk2rpmb: mmc2:0001 S40004 partition 3 4.00 MiB, chardev (2
42:0)
[    3.948480] asoc-audio-graph-card sound: wm8960-hifi-Playback <-> 4400a004
.audio-controller mapping ok
[    3.956775]  mmcblk2: p1 p2 p3
[    3.964691] asoc-audio-graph-card sound: wm8960-hifi-Capture <-> 4400a024.
audio-controller mapping ok
[    4.095274] input: joystick as /devices/platform/joystick/input/input1
```

```
[    4.101008] stm32_rtc 5c004000.rtc: setting system clock to 2000-01-01T00:
02:59 UTC (946684979)
[    4.110425] cfg80211: Loading compiled-in X.509 certificates for regulator
y database
[    4.119848] cfg80211: Loaded X.509 cert 'sforshee: 00b28ddf47aef9cea7'
[    4.125141] platform regulatory.0: Direct firmware load for regulatory.db
failed with error -2
[    4.134768] ALSA device list:
[    4.136471]   #0: WM8960-100ASK
[    4.139675] cfg80211: failed to load regulatory.db
[    4.419833] EXT4-fs (mmcblk2p3): recovery complete
[    4.424254] EXT4-fs (mmcblk2p3): mounted filesystem with ordered data mode
. Opts: (null)
[    4.431415] VFS: Mounted root (ext4 filesystem) on device 179:3.
[    4.438091] devtmpfs: mounted
[    4.441989] Freeing unused kernel memory: 1024K
[    4.445407] Run /sbin/init as init process
INIT: version  booting
[    4.681530] EXT4-fs (mmcblk2p3): re-mounted. Opts: (null)
INIT: Entering runlevel: 3
Starting syslogd: OK
Starting klogd: OK
Running sysctl: OK
Populating /dev using udev: [    5.269532] udevd[211]: starting version 3.2.9
[    5.351315] udevd[212]: starting eudev-3.2.9
[    5.939366] stm32_cec: Unknown symbol __stack_chk_guard (err -2)
[    6.262998] btintel: Unknown symbol __stack_chk_guard (err -2)
[    6.474148] galcore: loading out-of-tree module taints kernel.
[    6.722731] galcore: Unknown symbol __stack_chk_guard (err -2)
[    6.821749] bcmdhd: Unknown symbol __stack_chk_guard (err -2)
[    6.869090] bcmdhd: Unknown symbol __stack_chk_guard (err -2)
done
Initializing random number generator: OK
Saving random seed: OK
Starting system message bus: done
Starting rpcbind: [    7.215120] nf_defrag_ipv6: Unknown symbol __stack_chk_g
uard (err -2)
[    7.298578] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
OK
Starting network: [    7.374363] nf_defrag_ipv6: Unknown symbol __stack_chk_g
uard (err -2)
[    7.597298] stm32-dwmac 5800a000.ethernet eth0: PHY [stmmac-0:06] driver [
Generic PHY]
[    7.625419] dwmac4: Master AXI performs any burst length
[    7.629318] stm32-dwmac 5800a000.ethernet eth0: No Safety Features support
 found
[    7.636671] stm32-dwmac 5800a000.ethernet eth0: IEEE 1588-2008 Advanced Ti
mestamp supported
[    7.645650] stm32-dwmac 5800a000.ethernet eth0: registered PTP clock
[    7.651452] stm32-dwmac 5800a000.ethernet eth0: configuring for phy/rgmii
link mode
OK
Starting mosquitto: OK
Starting pulseaudio: [    7.936072] nf_defrag_ipv6: Unknown symbol __stack_ch
```

```
k_guard (err -2)
[    8.018779] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
OK
Starting sshd: [    8.286720] nf_defrag_ipv6: Unknown symbol __stack_chk_guar
d (err -2)
OK
[    8.443656] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
Starting NFS statd: [    8.556462] nf_defrag_ipv6: Unknown symbol __stack_chk
_guard (err -2)
[    8.628291] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
OK
Starting NFS services: OK
Starting NFS daemon: rpc.nfsd: Unable to access /proc/fs/nfsd errno 2 (No suc
h file or directory).
Please try, as root, 'mount -t nfsd nfsd /proc/fs/nfsd' and then restart rpc.
nfsd to correct the problem
FAIL
Starting NFS mountd: [    8.786287] nf_defrag_ipv6: Unknown symbol __stack_ch
k_guard (err -2)
rpc.mountd: svc_tli_create: could not open connection for udp6
[    8.848659] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
rpc.mountd: svc_tli_create: could not open connection for tcp6
[    8.920404] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
rpc.mountd: svc_tli_create: could not open connection for udp6
[    8.978166] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
rpc.mountd: svc_tli_create: could not open connection for tcp6
[    9.040808] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
rpc.mountd: svc_tli_create: could not open connection for udp6
[    9.108016] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
rpc.mountd: svc_tli_create: could not open connection for tcp6
OK
[    9.256350] file system registered
[    9.327361] read descriptors
[    9.328892] read strings
[    9.770611] stm32-dwmac 5800a000.ethernet eth0: Link is Up - 100Mbps/Full
- flow control rx/tx
[   10.316507] dwc2 49000000.usb-otg: bound driver configfs-gadget
[   10.365993] led_drv: module verification failed: signature and/or required
 key missing - tainting kernel
[   10.374248] led_drv: Unknown symbol __stack_chk_guard (err -2)
insmod: ERROR: could not insert module /root/led_drv.ko: Unknown symbol in mo
dule

Welcome to Buildroot
100ask login: open /dev/100ask_led err
QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-root'
Failed to open framebuffer /dev/fb0 (No such file or directory)
linuxfb: Failed to initialize screen
qt.qpa.input: xkbcommon not available, not performing key mapping
no screens available, assuming 24-bit color
Cannot create window: no screens available
[   34.407986] vmmcsd_fixed: disabling

Welcome to Buildroot
```

```
100ask login:
Welcome to Buildroot
100ask login: root
root@100ask:~# vi /etc/network/interfaces
root@100ask:~# ping 192.168.145.128
PING 192.168.145.128 (192.168.145.128): 56 data bytes
64 bytes from 192.168.145.128: seq=0 ttl=64 time=4.755 ms
64 bytes from 192.168.145.128: seq=1 ttl=64 time=2.295 ms
64 bytes from 192.168.145.128: seq=2 ttl=64 time=2.353 ms
^C
--- 192.168.145.128 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 2.295/3.134/4.755 ms
root@100ask:~# cd /dev
root@100ask:/dev# ls
autofs           mmcblk2rpmb      tty10            tty5
block            null             tty11            tty50
bus              phy_dev          tty12            tty51
char             ppp              tty13            tty52
console          ptmx             tty14            tty53
cpu_dma_latency  ptp0             tty15            tty54
disk             pts              tty16            tty55
fd               ptyp0            tty17            tty56
full             ptyp1            tty18            tty57
fuse             ptyp2            tty19            tty58
gpiochip0        ptyp3            tty2             tty59
gpiochip1        ptyp4            tty20            tty6
gpiochip2        ptyp5            tty21            tty60
gpiochip3        ptyp6            tty22            tty61
gpiochip4        ptyp7            tty23            tty62
gpiochip5        ram0             tty24            tty63
gpiochip6        ram1             tty25            tty7
gpiochip7        ram10            tty26            tty8
gpiochip8        ram11            tty27            tty9
gpiochip9        ram12            tty28            ttySTM0
hwrng            ram13            tty29            ttySTM1
i2c-0            ram14            tty3             ttyp0
i2c-1            ram15            tty30            ttyp1
i2c-2            ram2             tty31            ttyp2
input            ram3             tty32            ttyp3
kmsg             ram4             tty33            ttyp4
log              ram5             tty34            ttyp5
loop-control     ram6             tty35            ttyp6
loop0            ram7             tty36            ttyp7
loop1            ram8             tty37            ubi_ctrl
loop2            ram9             tty38            urandom
loop3            random           tty39            usb-ffs
loop4            rfkill           tty4             vcs
loop5            rtc              tty40            vcs1
loop6            rtc0             tty41            vcsa
loop7            shm              tty42            vcsa1
mem              snd              tty43            vcsu
mmcblk2          stderr           tty44            vcsu1
mmcblk2boot0     stdin            tty45            watchdog
```

```
mmcblk2boot1     stdout           tty46            watchdog0
mmcblk2p1        tty              tty47            zero
mmcblk2p2        tty0             tty48
mmcblk2p3        tty1             tty49
root@100ask:/dev# ls mmcblk2p2
mmcblk2p2
root@100ask:/dev# ll mmcblk2p2
-bash: ll: command not found
root@100ask:/dev# ls -l mmcblk2p2
brw-rw----    1 root     disk      179,   2 Jan  1 00:03 mmcblk2p2
root@100ask:/dev# mount /dev/mmcblk2p2 /boot/
root@100ask:/dev# mv /boot/uImage /boot/uImage_bad
root@100ask:/dev# ls /boot/
boot.scr.uimg                        stm32mp157c-100ask-512d-lcd-v1.dtb
mmc0_extlinux                        stm32mp157c-100ask-512d-v1.dtb
mmc1_extlinux                        uImage_bad
splash.bmp                           uInitrd
stm32mp157c-100ask-512d-hdmi-v1.dtb
root@100ask:/dev# mv /boot/stm32mp157c-100ask-512d-v1.dtb /boot/stm32mp157c-1
sk-512d-v1.dtb_bad
root@100ask:/dev# ls /mnt
root@100ask:/dev# mount -t nfs -o nolock,vers=3 192.168.145.128:/nfsroot /mnt
root@100ask:/dev# mount -t nfs -o nolock,vers=3 192.168.145.128:/nfsroot /mnt
root@100ask:/dev# ls /mnt/
01_hello                         lib
1.txt                            stm32mp157c-100ask-512d-v1.dtb
2.txt                            uImage
3.txt
root@100ask:/dev# mkimage -l /mnt/uImage
Image Name:   Linux-5.4.31-g893d15a84
Created:      Wed Oct 21 11:38:18 2026
Image Type:   ARM Linux Kernel Image (uncompressed)
Data Size:    8171272 Bytes = 7979.76 KiB = 7.79 MiB
Load Address: 0c200040
Entry Point:  0c200040
root@100ask:/dev# mkimage -l /mnt/uImage
Image Name:   Linux-5.4.31
Created:      Thu Oct 27 11:09:36 2022
Image Type:   ARM Linux Kernel Image (uncompressed)
Data Size:    8187608 Bytes = 7995.71 KiB = 7.81 MiB
Load Address: c2000040
Entry Point:  c2000040
root@100ask:/dev# cd ~
root@100ask:~# cp /mnt/uImage /boot/
root@100ask:~# cp /mnt/stm32mp157c-100ask-512d-v1.dtb  /boot/
root@100ask:~# sync
root@100ask:~# umount /boot
root@100ask:~# reboot

Broadcast message from root@100ask (console) (Sat Jan  1 03:11:40 2000):

The system is going down for reboot NOW!
INIT: Switching to runlevel: 6
root@100ask:~# killall: LED_and_TempHumi: no process killed
```

```
No process in pidfile '/var/run/adbd.pid' found running; none killed.
umount: /dev/usb-ffs/adb: target is busy.
Shutting down NFS mountd: OK
Shutting down NFS daemon: FAIL
Shutting down NFS services: OK
Stopping NFS statd: OK
Stopping sshd: OK
Stopping pulseaudio: OK
Stopping mosquitto: OK
Stopping network: [11325.577197] stm32-dwmac 5800a000.ethernet eth0: Link is
Down
OK
Stopping rpcbind daemon: OK
Stopping system message bus: done
Saving random seed: OK
Stopping klogd: OK
Stopping syslogd: OK
[11507.687970] nfs: server 192.168.145.128 not responding, still trying

root@100ask:~# umount /mnt

^C
root@100ask:~# poweroff
[12702.109393] reboot: Power down
INFO:    PSCI Power Domain Map:
INFO:      Domain Node : Level 1, parent_node -1, State ON (0x0)
INFO:      Domain Node : Level 0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0x0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0x1, parent_node 0, State ON (0x0)
 NOTICE:  CPU: STM32MP157AAC Rev.Z
NOTICE:  Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
INFO:    Reset reason (0x15):
INFO:      Power-on Reset (rst_por)
INFO:    Using EMMC
INFO:      Instance 2
INFO:    Boot used partition fsbl1
NOTICE:  BL2: v2.2-r1.0(debug):85b7ab8
NOTICE:  BL2: Built : 09:15:17, Aug 16 2026
INFO:    Using crypto library 'stm32_crypto_lib'
INFO:    BL2: Doing platform setup
INFO:    RAM: DDR3-1066/888 bin G 2x2Gb 533MHz v1.44
INFO:    Memory size = 0x20000000 (512 MB)
INFO:    BL2 runs SP_MIN setup
INFO:    BL2: Loading image id 4
INFO:    Loading image id=4 at address 0x2ffed000
INFO:    Image id=4 loaded: 0x2ffed000 - 0x2ffff000
INFO:    BL2: Loading image id 5
INFO:    Loading image id=5 at address 0xc0100000
INFO:    STM32 Image size : 869897
INFO:    Image id=5 loaded: 0xc0100000 - 0xc01d4609
WARNING: Skip signature check (header option)
NOTICE:  ROTPK is not deployed on platform. Skipping ROTPK verification.
NOTICE:  BL2: Booting BL32
INFO:    Entry point address = 0x2ffed000
```

```
INFO:    SPSR = 0x1d3
NOTICE:  SP_MIN: v2.2-r1.0(debug):85b7ab8
NOTICE:  SP_MIN: Built : 09:15:17, Aug 16 2026
INFO:    ARM GICv2 driver initialized
INFO:    stm32mp IWDG1 (12): Secure
INFO:    ETZPC: CRYP1 (9) could be non secure
INFO:    SP_MIN: Initializing runtime services
INFO:    SP_MIN: Preparing exit to normal world


U-Boot 2020.01-stm32mp-r1 (Aug 16 2026 - 09:15:35 +0800)

CPU: STM32MP157AAC Rev.Z
Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
Board: stm32mp1 in trusted mode (st,stm32mp157c-100ask-512d-v1)
DRAM:  512 MiB
Clocks:
- MPU : 650 MHz
- MCU : 208.878 MHz
- AXI : 266.500 MHz
- PER : 24 MHz
- DDR : 533 MHz
WDT:   Started with servicing (32s timeout)
NAND:  0 MiB
MMC:   STM32 SD/MMC: 0, STM32 SD/MMC: 1
Loading Environment from MMC... OK
In:    serial
Out:   serial
Err:   serial
Net:   eth0: ethernet@5800a000
Press SPACE to abort autoboot in 1 seconds
Boot over mmc1!
switch to partitions #0, OK
mmc1(part 0) is current device
Scanning mmc 1:2...
Found /mmc1_extlinux/stm32mp157c-100ask-512d-v1_extlinux.conf
Retrieving file: /mmc1_extlinux/stm32mp157c-100ask-512d-v1_extlinux.conf
210 bytes read in 38 ms (4.9 KiB/s)
1:      100ask-lcd
Retrieving file: /uImage
8187672 bytes read in 317 ms (24.6 MiB/s)
append: root=/dev/mmcblk2p3 rootwait rw console=ttySTM0,115200
Retrieving file: /stm32mp157c-100ask-512d-lcd-v1.dtb
74438 bytes read in 40 ms (1.8 MiB/s)
## Booting kernel from Legacy Image at c2000000 ...
   Image Name:   Linux-5.4.31
   Created:      2022-10-27  11:09:36 UTC
   Image Type:   ARM Linux Kernel Image (uncompressed)
   Data Size:    8187608 Bytes = 7.8 MiB
   Load Address: c2000040
   Entry Point:  c2000040
   Verifying Checksum ... OK
## Flattened Device Tree blob at c4000000
   Booting using the fdt blob at 0xc4000000
```

```
   XIP Kernel Image
   Loading Device Tree to cffea000, end cffff2c5 ... OK

Starting kernel ...

[    0.000000] Booting Linux on physical CPU 0x0
[    0.000000] Linux version 5.4.31 (book@virtual-machine) (gcc version 8.4.0 (
Buildroot 2020.02-g8ad4a8c-dirty)) #1 SMP PREEMPT Thu Oct 27 11:04:26 UTC 2022
[    0.000000] CPU: ARMv7 Processor [410fc075] revision 5 (ARMv7), cr=10c5387d
[    0.000000] CPU: div instructions available: patching division code
[    0.000000] CPU: PIPT / VIPT nonaliasing data cache, VIPT aliasing instructi
on cache
[    0.000000] OF: fdt: Machine model: 100ASK YA157C v2 www.100ask.com
[    0.000000] Memory policy: Data cache writealloc
[    0.000000] Reserved memory: created DMA memory pool at 0x10000000, size 0 M
iB
[    0.000000] OF: reserved mem: initialized node mcuram2@10000000, compatible
id shared-dma-pool
[    0.000000] Reserved memory: created DMA memory pool at 0x10040000, size 0 M
iB
[    0.000000] OF: reserved mem: initialized node vdev0vring0@10040000, compati
ble id shared-dma-pool
[    0.000000] Reserved memory: created DMA memory pool at 0x10041000, size 0 M
iB
[    0.000000] OF: reserved mem: initialized node vdev0vring1@10041000, compati
ble id shared-dma-pool
[    0.000000] Reserved memory: created DMA memory pool at 0x10042000, size 0 M
iB
[    0.000000] OF: reserved mem: initialized node vdev0buffer@10042000, compati
ble id shared-dma-pool
[    0.000000] Reserved memory: created DMA memory pool at 0x30000000, size 0 M
iB
[    0.000000] OF: reserved mem: initialized node mcuram@30000000, compatible i
d shared-dma-pool
[    0.000000] Reserved memory: created DMA memory pool at 0x38000000, size 0 M
iB
[    0.000000] OF: reserved mem: initialized node retram@38000000, compatible i
d shared-dma-pool
[    0.000000] cma: Reserved 128 MiB at 0xd2000000
[    0.000000] psci: probing for conduit method from DT.
[    0.000000] psci: PSCIv1.1 detected in firmware.
[    0.000000] psci: Using standard PSCI v0.2 function IDs
[    0.000000] psci: MIGRATE_INFO_TYPE not supported.
[    0.000000] psci: SMC Calling Convention v1.0
[    0.000000] percpu: Embedded 20 pages/cpu s50124 r8192 d23604 u81920
[    0.000000] Built 1 zonelists, mobility grouping on.  Total pages: 113536
[    0.000000] Kernel command line: root=/dev/mmcblk2p3 rootwait rw console=tty
STM0,115200
[    0.000000] Dentry cache hash table entries: 65536 (order: 6, 262144 bytes,
linear)
[    0.000000] Inode-cache hash table entries: 32768 (order: 5, 131072 bytes, l
inear)
[    0.000000] mem auto-init: stack:off, heap alloc:off, heap free:off
[    0.000000] Memory: 303880K/458752K available (12288K kernel code, 962K rwda
```

```
ta, 3516K rodata, 1024K init, 226K bss, 23800K reserved, 131072K cma-reserved,
0K highmem)
[    0.000000] SLUB: HWalign=64, Order=0-3, MinObjects=0, CPUs=2, Nodes=1
[    0.000000] rcu: Preemptible hierarchical RCU implementation.
[    0.000000] rcu:     RCU event tracing is enabled.
[    0.000000]  Tasks RCU enabled.
[    0.000000] rcu: RCU calculated value of scheduler-enlistment delay is 10 ji
ffies.
[    0.000000] NR_IRQS: 16, nr_irqs: 16, preallocated irqs: 16
[    0.000000] random: get_random_bytes called from start_kernel+0x320/0x4f4 wi
th crng_init=0
[    0.000000] arch_timer: cp15 timer(s) running at 24.00MHz (virt).
[    0.000000] clocksource: arch_sys_counter: mask: 0xffffffffffffff max_cycles
: 0x588fe9dc0, max_idle_ns: 440795202592 ns
[    0.000009] sched_clock: 56 bits at 24MHz, resolution 41ns, wraps every 4398
046511097ns
[    0.000027] Switching to timer-based delay loop, resolution 41ns
[    0.001867] Console: colour dummy device 80x30
[    0.001919] Calibrating delay loop (skipped), value calculated using timer f
requency.. 48.00 BogoMIPS (lpj=240000)
[    0.001939] pid_max: default: 32768 minimum: 301
[    0.002200] Mount-cache hash table entries: 1024 (order: 0, 4096 bytes, line
ar)
[    0.002224] Mountpoint-cache hash table entries: 1024 (order: 0, 4096 bytes,
 linear)
[    0.003515] CPU: Testing write buffer coherency: ok
[    0.004048] /cpus/cpu@0 missing clock-frequency property
[    0.004082] /cpus/cpu@1 missing clock-frequency property
[    0.004101] CPU0: thread -1, cpu 0, socket 0, mpidr 80000000
[    0.059801] Setting up static identity map for 0xc0100000 - 0xc0100060
[    0.079774] rcu: Hierarchical SRCU implementation.
[    0.105149] smp: Bringing up secondary CPUs ...
[    0.180418] CPU1: thread -1, cpu 1, socket 0, mpidr 80000001
[    0.180638] smp: Brought up 1 node, 2 CPUs
[    0.180662] SMP: Total of 2 processors activated (96.00 BogoMIPS).
[    0.180673] CPU: All CPU(s) started in SVC mode.
[    0.181531] devtmpfs: initialized
[    0.209892] VFP support v0.3: implementor 41 architecture 2 part 30 variant
7 rev 5
[    0.210589] clocksource: jiffies: mask: 0xffffffff max_cycles: 0xffffffff, m
ax_idle_ns: 19112604462750000 ns
[    0.210626] futex hash table entries: 512 (order: 3, 32768 bytes, linear)
[    0.217304] pinctrl core: initialized pinctrl subsystem
[    0.219664] NET: Registered protocol family 16
[    0.233140] DMA: preallocated 256 KiB pool for atomic coherent allocations
[    0.235682] hw-breakpoint: found 5 (+1 reserved) breakpoint and 4 watchpoint
 registers.
[    0.235704] hw-breakpoint: maximum watchpoint size is 8 bytes.
[    0.236128] Serial: AMBA PL011 UART driver
[    0.239260] stm32-pm-domain pm_domain: domain core-ret-power-domain register
ed
[    0.239320] stm32-pm-domain pm_domain: subdomain core-power-domain registere
d
[    0.239337] stm32-pm-domain pm_domain: domains probed
```

```
[    0.291979] gpio-regulator regulators: No 'states' property found
[    0.292020] gpio-regulator: probe of regulators failed with error -22
[    0.293408] SCSI subsystem initialized
[    0.294274] usbcore: registered new interface driver usbfs
[    0.294357] usbcore: registered new interface driver hub
[    0.294506] usbcore: registered new device driver usb
[    0.294892] pps_core: LinuxPPS API ver. 1 registered
[    0.294906] pps_core: Software ver. 5.3.6 - Copyright 2005-2007 Rodolfo Giom
etti <giometti@linux.it>
[    0.294949] PTP clock support registered
[    0.295315] EDAC MC: Ver: 3.0.0
[    0.296647] Advanced Linux Sound Architecture Driver Initialized.
[    0.297526] Bluetooth: Core ver 2.22
[    0.297613] NET: Registered protocol family 31
[    0.297626] Bluetooth: HCI device and connection manager initialized
[    0.297649] Bluetooth: HCI socket layer initialized
[    0.297666] Bluetooth: L2CAP socket layer initialized
[    0.297703] Bluetooth: SCO socket layer initialized
[    0.298682] clocksource: Switched to clocksource arch_sys_counter
[    1.058226] thermal_sys: Registered thermal governor 'step_wise'
[    1.058952] NET: Registered protocol family 2
[    1.060078] tcp_listen_portaddr_hash hash table entries: 512 (order: 0, 6144
 bytes, linear)
[    1.060134] TCP established hash table entries: 4096 (order: 2, 16384 bytes,
 linear)
[    1.060213] TCP bind hash table entries: 4096 (order: 3, 32768 bytes, linear
)
[    1.060310] TCP: Hash tables configured (established 4096 bind 4096)
[    1.060458] UDP hash table entries: 256 (order: 1, 8192 bytes, linear)
[    1.060514] UDP-Lite hash table entries: 256 (order: 1, 8192 bytes, linear)
[    1.060843] NET: Registered protocol family 1
[    1.061797] RPC: Registered named UNIX socket transport module.
[    1.061818] RPC: Registered udp transport module.
[    1.061828] RPC: Registered tcp transport module.
[    1.061838] RPC: Registered tcp NFSv4.1 backchannel transport module.
[    1.063717] hw perfevents: enabled with armv7_cortex_a7 PMU driver, 5 counte
rs available
[    1.065817] Initialise system trusted keyrings
[    1.066184] workingset: timestamp_bits=14 max_order=17 bucket_order=3
[    1.078918] squashfs: version 4.0 (2009/01/31) Phillip Lougher
[    1.090367] NFS: Registering the id_resolver key type
[    1.090429] Key type id_resolver registered
[    1.090441] Key type id_legacy registered
[    1.090665] ntfs: driver 2.1.32 [Flags: R/O].
[    1.091169] jffs2: version 2.2. (NAND) © 2001-2006 Red Hat, Inc.
[    1.091963] fuse: init (API version 7.31)
[    1.146877] NET: Registered protocol family 38
[    1.146907] Key type asymmetric registered
[    1.146921] Asymmetric key parser 'x509' registered
[    1.147014] Block layer SCSI generic (bsg) driver version 0.4 loaded (major
246)
[    1.147032] io scheduler mq-deadline registered
[    1.147044] io scheduler kyber registered
[    1.157522] STM32 USART driver initialized
```

```
[    1.159311] ar8035 phy init
[    1.159787] the phy gpio failed
[    1.181326] brd: module loaded
[    1.195717] loop: module loaded
[    1.202226] libphy: Fixed MDIO Bus: probed
[    1.204097] CAN device driver interface
[    1.206658] PPP generic driver version 2.4.2
[    1.207011] PPP BSD Compression module registered
[    1.207027] PPP Deflate Compression module registered
[    1.207053] PPP MPPE Compression module registered
[    1.207067] NET: Registered protocol family 24
[    1.207113] SLIP: version 0.8.4-NET3.019-NEWTTY (dynamic channels, max=256)
(6 bit encapsulation enabled).
[    1.207124] CSLIP: code copyright 1989 Regents of the University of Californ
ia.
[    1.207133] SLIP linefill/keepalive option.
[    1.207147] pegasus: v0.9.3 (2013/04/25), Pegasus/Pegasus II USB Ethernet dr
iver
[    1.207252] usbcore: registered new interface driver pegasus
[    1.207334] usbcore: registered new interface driver r8152
[    1.207408] usbcore: registered new interface driver asix
[    1.207485] usbcore: registered new interface driver ax88179_178a
[    1.207547] usbcore: registered new interface driver cdc_ether
[    1.207624] usbcore: registered new interface driver smsc75xx
[    1.207703] usbcore: registered new interface driver smsc95xx
[    1.207762] usbcore: registered new interface driver net1080
[    1.207821] usbcore: registered new interface driver cdc_subset
[    1.207890] usbcore: registered new interface driver zaurus
[    1.207981] usbcore: registered new interface driver cdc_ncm
[    1.209962] ehci_hcd: USB 2.0 'Enhanced' Host Controller (EHCI) Driver
[    1.210023] ehci-platform: EHCI generic platform driver
[    1.210546] ohci_hcd: USB 1.1 'Open' Host Controller (OHCI) Driver
[    1.210594] ohci-platform: OHCI generic platform driver
[    1.211647] usbcore: registered new interface driver cdc_acm
[    1.211663] cdc_acm: USB Abstract Control Model driver for USB modems and IS
DN adapters
[    1.211747] usbcore: registered new interface driver usblp
[    1.211809] usbcore: registered new interface driver cdc_wdm
[    1.211868] usbcore: registered new interface driver uas
[    1.211949] usbcore: registered new interface driver usb-storage
[    1.212121] usbcore: registered new interface driver usbserial_generic
[    1.212173] usbserial: USB Serial support registered for generic
[    1.212231] usbcore: registered new interface driver option
[    1.212273] usbserial: USB Serial support registered for GSM modem (1-port)
[    1.217412] i2c /dev entries driver
[    1.222653] stm32-cpufreq stm32-cpufreq: Failed to get chip info: -517
[    1.226972] sdhci: Secure Digital Host Controller Interface driver
[    1.226988] sdhci: Copyright(c) Pierre Ossman
[    1.226998] Synopsys Designware Multimedia Card Interface Driver
[    1.227601] sdhci-pltfm: SDHCI platform and OF driver helper
[    1.229018] ledtrig-cpu: registered to indicate activity on CPUs
[    1.231486] usbcore: registered new interface driver usbhid
[    1.231505] usbhid: USB HID core driver
[    1.231516] dtbocfg_module_init
```

```
[    1.231570] dtbocfg_module_init: OK
[    1.231934] arm-smc-mbox mailbox-0: ARM SMC mailbox enabled.
[    1.232070] arm-smc-mbox mailbox-1: ARM SMC mailbox enabled.
[    1.233834] remoteproc remoteproc0: releasing m4
[    1.237092] Unable to get STM32 DDR PMU clock
[    1.242617] drop_monitor: Initializing network drop monitor service
[    1.242959] NET: Registered protocol family 17
[    1.242993] can: controller area network core (rev 20170425 abi 9)
[    1.243124] NET: Registered protocol family 29
[    1.243141] can: raw protocol (rev 20170425)
[    1.243154] can: broadcast manager protocol (rev 20170425 t)
[    1.243172] can: netlink gateway (rev 20190810) max_hops=1
[    1.243712] Bluetooth: RFCOMM TTY layer initialized
[    1.243758] Bluetooth: RFCOMM socket layer initialized
[    1.243810] Bluetooth: RFCOMM ver 1.11
[    1.243871] Bluetooth: BNEP (Ethernet Emulation) ver 1.3
[    1.243885] Bluetooth: BNEP filters: protocol multicast
[    1.243906] Bluetooth: BNEP socket layer initialized
[    1.243919] Bluetooth: HIDP (Human Interface Emulation) ver 1.2
[    1.243937] Bluetooth: HIDP socket layer initialized
[    1.244525] Key type dns_resolver registered
[    1.244720] ThumbEE CPU extension supported.
[    1.244747] Registering SWP/SWPB emulation handler
[    1.245183] registered taskstats version 1
[    1.245198] Loading compiled-in X.509 certificates
[    1.256468] Loaded X.509 cert 'Build time autogenerated kernel key: a551a3bc
af9a62271059adf2e1003b2f3df06de0'
[    1.283400] arm-scmi firmware:scmi-0: SCMI Protocol v2.0 'ST:' Firmware vers
ion 0x0
[    1.287587] remoteproc remoteproc0: releasing m4
[    1.289246] Unable to get STM32 DDR PMU clock
[    1.312689] debugfs: Directory 'cpu0' with parent 'opp' already present!
[    1.322249] remoteproc remoteproc0: releasing m4
[    1.323713] Unable to get STM32 DDR PMU clock
[    1.362749] stm32-mdma 58000000.dma: STM32 MDMA driver registered
[    1.367256] stm32-dma 48000000.dma: STM32 DMA driver registered
[    1.369526] stm32-dma 48001000.dma: STM32 DMA driver registered
[    1.370207] reg11: supplied by regulator-dummy
[    1.370679] reg18: supplied by regulator-dummy
[    1.371121] usb33: supplied by regulator-dummy
[    1.389123] stm_thermal 50028000.thermal: stm_thermal_probe: Driver initiali
zed successfully
[    1.391892] debugfs: Directory 'cpu0' with parent 'opp' already present!
[    1.395907] random: fast init done
[    1.396053] stm32-crc32 58009000.crc: Initialized
[    1.397670] stm32-hash 54002000.hash: will run requests pump with realtime p
riority
[    1.400312] random: crng init done
[    1.402623] stm32-hash 54002000.hash: Init HASH done HW ver 23 DMA mode 1
[    1.415034] remoteproc remoteproc0: releasing m4
[    1.416828] stm32-ddr-pmu: probed (ID=0x00140061 VER=0x00000010), DDR@533MHz
[    1.419321] /soc/interrupt-controller@5000d000: bank0
[    1.419341] /soc/interrupt-controller@5000d000: bank1
[    1.419354] /soc/interrupt-controller@5000d000: bank2
```

```
[    1.421649] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOA bank added
[    1.422244] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOB bank added
[    1.422768] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOC bank added
[    1.423295] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOD bank added
[    1.423794] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOE bank added
[    1.424306] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOF bank added
[    1.424837] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOG bank added
[    1.425351] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOH bank added
[    1.425839] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOI bank added
[    1.426145] stm32mp157-pinctrl soc:pin-controller@50002000: Pinctrl STM32 in
itialized
[    1.427240] stm32mp157-pinctrl soc:pin-controller-z@54004000: GPIOZ bank add
ed
[    1.427270] stm32mp157-pinctrl soc:pin-controller-z@54004000: Pinctrl STM32
initialized
[    1.429813] stm32-usbphyc 5a006000.usbphyc: registered rev:1.0
[    1.431331] stm32-usart 40010000.serial: interrupt mode for rx (no dma)
[    1.431354] stm32-usart 40010000.serial: interrupt mode for tx (no dma)
[    1.431390] 40010000.serial: ttySTM0 at MMIO 0x40010000 (irq = 58, base_baud
 = 4000000) is a stm32-usart
[    2.814409] printk: console [ttySTM0] enabled
[    2.820679] 44003000.serial: ttySTM1 at MMIO 0x44003000 (irq = 59, base_baud
 = 4000000) is a stm32-usart
[    2.830344] stm32-usart 5c000000.serial: interrupt mode for rx (no dma)
[    2.835498] stm32-usart 5c000000.serial: interrupt mode for tx (no dma)
[    2.842200] 5c000000.serial: ttySTM2 at MMIO 0x5c000000 (irq = 60, base_baud
 = 4000000) is a stm32-usart
[    2.851867] serial serial0: tty port ttySTM2 registered
[    2.863472] inv-mpu6000-spi spi0.0: mounting matrix not found: using identit
y...
[    2.869566] inv-mpu6000-spi spi0.0: spi0.0 supply vddio not found, using dum
my regulator
[    3.092371] spi_stm32 44009000.spi: driver initialized
[    3.097822] m_can_platform 4400f000.can: m_can device registered (irq=55, ve
rsion=32)
[    3.106196] stm32-dwmac 5800a000.ethernet: PTP uses main clock
[    3.110647] stm32-dwmac 5800a000.ethernet: no reset control found
[    3.116687] stm32-dwmac 5800a000.ethernet: No phy clock provided...
[    3.123601] stm32-dwmac 5800a000.ethernet: User ID: 0x40, Synopsys ID: 0x42
[    3.129899] stm32-dwmac 5800a000.ethernet:   DWMAC4/5
[    3.134807] stm32-dwmac 5800a000.ethernet: DMA HW capability register suppor
ted
[    3.142130] stm32-dwmac 5800a000.ethernet: RX Checksum Offload Engine suppor
ted
[    3.149422] stm32-dwmac 5800a000.ethernet: TX Checksum insertion supported
[    3.156263] stm32-dwmac 5800a000.ethernet: Wake-Up On Lan supported
[    3.162553] stm32-dwmac 5800a000.ethernet: TSO supported
[    3.167818] stm32-dwmac 5800a000.ethernet: Enable RX Mitigation via HW Watch
dog Timer
[    3.175668] stm32-dwmac 5800a000.ethernet: TSO feature enabled
[    3.181907] libphy: stmmac: probed
[    3.185130] yzhang..get YT8511, abt to set 125m clk out, phyaddr=6, phyid=00
00010a
[    3.193425] yt8511_config_out_125m, phy clk out, val=0x008057
```

```
[    3.198508] yzhang..8511 set 125m clk out, reg=0x8057
[    3.205722] dwc2 49000000.usb-otg: 49000000.usb-otg supply vusb_d not found,
 using dummy regulator
[    3.213450] dwc2 49000000.usb-otg: 49000000.usb-otg supply vusb_a not found,
 using dummy regulator
[    3.348983] dwc2 49000000.usb-otg: EPs: 9, dedicated fifos, 952 entries in S
PRAM
[    3.355997] dwc2 49000000.usb-otg: DWC OTG Controller
[    3.360076] dwc2 49000000.usb-otg: new USB bus registered, assigned bus numb
er 1
[    3.367396] dwc2 49000000.usb-otg: irq 65, io mem 0x49000000
[    3.373366] usb usb1: New USB device found, idVendor=1d6b, idProduct=0002, b
cdDevice= 5.04
[    3.381283] usb usb1: New USB device strings: Mfr=3, Product=2, SerialNumber
=1
[    3.388457] usb usb1: Product: DWC OTG Controller
[    3.393172] usb usb1: Manufacturer: Linux 5.4.31 dwc2_hsotg
[    3.398736] usb usb1: SerialNumber: 49000000.usb-otg
[    3.404595] hub 1-0:1.0: USB hub found
[    3.407473] hub 1-0:1.0: 1 port detected
[    3.413115] ehci-platform 5800d000.usbh-ehci: EHCI Host Controller
[    3.417865] ehci-platform 5800d000.usbh-ehci: new USB bus registered, assign
ed bus number 2
[    3.426681] ehci-platform 5800d000.usbh-ehci: irq 66, io mem 0x5800d000
[    3.458727] ehci-platform 5800d000.usbh-ehci: USB 2.0 started, EHCI 1.00
[    3.464385] usb usb2: New USB device found, idVendor=1d6b, idProduct=0002, b
cdDevice= 5.04
[    3.472267] usb usb2: New USB device strings: Mfr=3, Product=2, SerialNumber
=1
[    3.479451] usb usb2: Product: EHCI Host Controller
[    3.484294] usb usb2: Manufacturer: Linux 5.4.31 ehci_hcd
[    3.489707] usb usb2: SerialNumber: 5800d000.usbh-ehci
[    3.495756] hub 2-0:1.0: USB hub found
[    3.498611] hub 2-0:1.0: 2 ports detected
[    3.504025] ohci-platform 5800c000.usbh-ohci: Generic Platform OHCI controll
er
[    3.509896] ohci-platform 5800c000.usbh-ohci: new USB bus registered, assign
ed bus number 3
[    3.518546] ohci-platform 5800c000.usbh-ohci: irq 52, io mem 0x5800c000
[    3.593040] usb usb3: New USB device found, idVendor=1d6b, idProduct=0001, b
cdDevice= 5.04
[    3.599887] usb usb3: New USB device strings: Mfr=3, Product=2, SerialNumber
=1
[    3.607054] usb usb3: Product: Generic Platform OHCI controller
[    3.612989] usb usb3: Manufacturer: Linux 5.4.31 ohci_hcd
[    3.618349] usb usb3: SerialNumber: 5800c000.usbh-ohci
[    3.625270] hub 3-0:1.0: USB hub found
[    3.627615] hub 3-0:1.0: 2 ports detected
[    3.634413] stm32_rtc 5c004000.rtc: registered as rtc0
[    3.638936] stm32_rtc 5c004000.rtc: Date/Time must be initialized
[    3.644174] stm32_rtc 5c004000.rtc: registered rev:1.2
[    3.670374] stm32f7-i2c 40012000.i2c: doesn't use DMA
[    3.676118] fusb302 0-0022: Can't get property of role, set role to default
DRP
```

```
[    3.689941] fusb302 0-0022: port 0 probe success with role ROLE_MODE_DRP, tr
y_role ROLE_MODE_NONE
[    3.698114] stm32f7-i2c 40012000.i2c: STM32F7 I2C-0 bus adapter
[    3.724496] stm32f7-i2c 40013000.i2c: doesn't use DMA
[    3.728848] stm32f7-i2c 40013000.i2c: STM32F7 I2C-1 bus adapter
[    3.755186] stm32f7-i2c 5c002000.i2c: doesn't use DMA
[    3.760227] Goodix-TS 2-005d: 2-005d supply AVDD28 not found, using dummy re
gulator
[    3.766579] Goodix-TS 2-005d: 2-005d supply VDDIO not found, using dummy reg
ulator
[    3.890180] Goodix-TS 2-005d: ID 911, version: 1060
[    3.894430] Goodix-TS 2-005d: Direct firmware load for goodix_911_cfg.bin fa
iled with error -2
[    3.919359] input: Goodix Capacitive TouchScreen as /devices/platform/soc/5c
002000.i2c/i2c-2/2-005d/input/input0
[    3.932072] stm32f7-i2c 5c002000.i2c: STM32F7 I2C-2 bus adapter
[    3.939178] mmci-pl18x 48004000.sdmmc: allocated mmc-pwrseq
[    3.943722] mmci-pl18x 48004000.sdmmc: mmc0: PL180 manf 53 rev1 at 0x4800400
0 irq 43,0 (pio)
[    3.979922] mmci-pl18x 58005000.sdmmc: mmc1: PL180 manf 53 rev2 at 0x5800500
0 irq 49,0 (pio)
[    4.003888] mmc0: queuing unknown CIS tuple 0x80 (2 bytes)
[    4.009361] mmc0: queuing unknown CIS tuple 0x80 (3 bytes)
[    4.015530] mmci-pl18x 58007000.sdmmc: mmc2: PL180 manf 53 rev2 at 0x5800700
0 irq 50,0 (pio)
[    4.024801] mmc0: queuing unknown CIS tuple 0x80 (3 bytes)
[    4.032557] mmc0: queuing unknown CIS tuple 0x80 (7 bytes)
[    4.042077] mmc0: queuing unknown CIS tuple 0x81 (9 bytes)
[    4.054035] stm32-ipcc 4c001000.mailbox: ipcc rev:1.0 enabled, 6 chans, proc
 0
[    4.063842] stm32-rproc mlahb:m4@10000000: wdg irq registered
[    4.070318] remoteproc remoteproc0: m4 is available
[    4.075850] stm32mp157-pinctrl soc:pin-controller@50002000: missing pins pro
perty in node pins .
[    4.083348] stm32-adc-core: probe of 48003000.adc failed with error -22
[    4.101715] [drm] Supports vblank timestamp caching Rev 2 (21.10.2013).
[    4.106866] [drm] Driver supports precise vblank timestamp query.
[    4.117120] [drm] Initialized stm 1.0.0 20170330 for 5a001000.display-contro
ller on minor 0
[    4.178293] mmc2: new high speed MMC card at address 0001
[    4.189953] mmcblk2: mmc2:0001 S40004 3.64 GiB
[    4.190645] mmcblk2boot0: mmc2:0001 S40004 partition 1 4.00 MiB
[    4.191328] mmcblk2boot1: mmc2:0001 S40004 partition 2 4.00 MiB
[    4.192894] mmcblk2rpmb: mmc2:0001 S40004 partition 3 4.00 MiB, chardev (242
:0)
[    4.204649] mmc0: new high speed SDIO card at address 0001
[    4.223358]  mmcblk2: p1 p2 p3
[    4.289023] Console: switching to colour frame buffer device 128x37
[    4.348383] stm32-display 5a001000.display-controller: fb0: stmdrmfb frame b
uffer device
[    4.488199] asoc-audio-graph-card sound: wm8960-hifi-Playback <-> 4400a004.a
udio-controller mapping ok
[    4.497231] asoc-audio-graph-card sound: wm8960-hifi-Capture <-> 4400a024.au
dio-controller mapping ok
```

```
[    4.626061] input: joystick as /devices/platform/joystick/input/input1
[    4.631771] stm32_rtc 5c004000.rtc: setting system clock to 2000-01-01T00:00
:09 UTC (946684809)
[    4.641020] cfg80211: Loading compiled-in X.509 certificates for regulatory
database
[    4.650636] cfg80211: Loaded X.509 cert 'sforshee: 00b28ddf47aef9cea7'
[    4.655928] platform regulatory.0: Direct firmware load for regulatory.db fa
iled with error -2
[    4.658593] ALSA device list:
[    4.664429] cfg80211: failed to load regulatory.db
[    4.667257]   #0: WM8960-100ASK
[    4.881547] EXT4-fs (mmcblk2p3): recovery complete
[    4.885950] EXT4-fs (mmcblk2p3): mounted filesystem with ordered data mode.
Opts: (null)
[    4.893133] VFS: Mounted root (ext4 filesystem) on device 179:3.
[    4.899766] devtmpfs: mounted
[    4.903684] Freeing unused kernel memory: 1024K
[    4.907102] Run /sbin/init as init process
INIT: version  booting
[    5.141591] EXT4-fs (mmcblk2p3): re-mounted. Opts: (null)
INIT: Entering runlevel: 3
Starting syslogd: OK
Starting klogd: OK
Running sysctl: OK
Populating /dev using udev: [    5.729601] udevd[215]: starting version 3.2.9
[    5.811028] udevd[216]: starting eudev-3.2.9
[    6.363063] stm32_cec: Unknown symbol __stack_chk_guard (err -2)
[    6.723238] btintel: Unknown symbol __stack_chk_guard (err -2)
[    7.067509] galcore: loading out-of-tree module taints kernel.
[    7.254332] galcore: Unknown symbol __stack_chk_guard (err -2)
[    7.260284] bcmdhd: Unknown symbol __stack_chk_guard (err -2)
[    7.371181] bcmdhd: Unknown symbol __stack_chk_guard (err -2)
done
Initializing random number generator: OK
Saving random seed: OK
Starting system message bus: done
Starting rpcbind: [    7.731545] nf_defrag_ipv6: Unknown symbol __stack_chk_gua
rd (err -2)
[    7.801907] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
OK
[    7.873945] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
Starting network: [    8.093333] stm32-dwmac 5800a000.ethernet eth0: PHY [stmma
c-0:06] driver [Generic PHY]
[    8.111396] dwmac4: Master AXI performs any burst length
[    8.115273] stm32-dwmac 5800a000.ethernet eth0: No Safety Features support f
ound
[    8.122853] stm32-dwmac 5800a000.ethernet eth0: IEEE 1588-2008 Advanced Time
stamp supported
[    8.131501] stm32-dwmac 5800a000.ethernet eth0: registered PTP clock
[    8.137348] stm32-dwmac 5800a000.ethernet eth0: configuring for phy/rgmii li
nk mode
[    8.148043] stm32-dwmac 5800a000.ethernet eth0: Link is Up - 100Mbps/Full -
flow control rx/tx
OK
```

```
Starting mosquitto: OK
Starting pulseaudio: [    8.413506] nf_defrag_ipv6: Unknown symbol __stack_chk_
guard (err -2)
[    8.480342] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
OK
Starting sshd: [    8.765686] nf_defrag_ipv6: Unknown symbol __stack_chk_guard
(err -2)
OK
[    8.929198] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
Starting NFS statd: [    9.049423] nf_defrag_ipv6: Unknown symbol __stack_chk_g
uard (err -2)
[    9.118794] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
OK
Starting NFS services: OK
Starting NFS daemon: rpc.nfsd: Unable to access /proc/fs/nfsd errno 2 (No such
file or directory).
Please try, as root, 'mount -t nfsd nfsd /proc/fs/nfsd' and then restart rpc.nf
sd to correct the problem
FAIL
Starting NFS mountd: [    9.273310] nf_defrag_ipv6: Unknown symbol __stack_chk_
guard (err -2)
rpc.mountd: svc_tli_create: could not open connection for udp6
[    9.359874] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
rpc.mountd: svc_tli_create: could not open connection for tcp6
[    9.401277] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
rpc.mountd: svc_tli_create: could not open connection for udp6
[    9.459107] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
rpc.mountd: svc_tli_create: could not open connection for tcp6
[    9.531486] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
rpc.mountd: svc_tli_create: could not open connection for udp6
[    9.589040] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
rpc.mountd: svc_tli_create: could not open connection for tcp6
OK
[    9.753176] file system registered
[    9.824318] read descriptors
[    9.825755] read strings
[   10.814073] dwc2 49000000.usb-otg: bound driver configfs-gadget
[   10.864441] led_drv: module verification failed: signature and/or required k
ey missing - tainting kernel
[   10.872665] led_drv: Unknown symbol __stack_chk_guard (err -2)
insmod: ERROR: could not insert module /root/led_drv.ko: Unknown symbol in modu
le

Welcome to Buildroot
100ask login: open /dev/100ask_led err
QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-root'
qt.qpa.input: xkbcommon not available, not performing key mapping
[   35.048754] vmmcsd_fixed: disabling

Welcome to Buildroot
100ask login: root
root@100ask:~# poweroff

Broadcast message from root@100ask (console) (Sat Jan  1 00:01:18 2000):
```

```
The system is going down for system halt NOW!
INIT: Switching to runlevel: 0
root@100ask:~# No process in pidfile '/var/run/adbd.pid' found running; none ki
lled.
umount: /dev/usb-ffs/adb: target is busy.
Shutting down NFS mountd: OK
Shutting down NFS daemon: FAIL
Shutting down NFS services: OK
Stopping NFS statd: OK
Stopping sshd: OK
Stopping pulseaudio: OK
Stopping mosquitto: OK
Stopping network: [   73.852398] stm32-dwmac 5800a000.ethernet eth0: Link is Do
wn
OK
Stopping rpcbind daemon: OK
Stopping system message bus: done
Saving random seed: OK
Stopping klogd: OK
Stopping syslogd: OK
umount: /dev/usb-ffs/adb: target is busy.
[   74.258345] EXT4-fs (mmcblk2p3): re-mounted. Opts: (null)
[   74.312173] reboot: Power down
INFO:    PSCI Power Domain Map:
INFO:      Domain Node : Level 1, parent_node -1, State ON (0x0)
INFO:      Domain Node : Level 0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0x0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0x1, parent_node 0, State ON (0x0)
```